

Chapter 10

Straight Lines

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Distance Formula and Section Formula]

1. The distance between the point $(\sin\theta, 0)$ and $(0, \cos\theta)$ is

(1) 1 (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) $\frac{1}{4}$

Sol. Answer (1)

$$\text{Distance} = \sqrt{\sin^2 \theta + \cos^2 \theta} = 1$$

2. The coordinates of the centroid of triangle formed by the vertices $(1, 1)$, $(2, 6)$, $(6, -1)$ is

(1) $(3, 1)$ (2) $(3, 2)$ (3) $(2, 4)$ (4) $(4, 6)$

Sol. Answer (2)

$$\text{Centroid} = \left(\frac{1+2+6}{3}, \frac{1+6-1}{3} \right) = (3, 2)$$

3. If the area of the triangle formed by the points $(0, 0)$, $(h, 0)$ and $(0, 4)$ is 2, then the sum of values of h is

(1) 1 (2) 2 (3) 0 (4) 4

Sol. Answer (3)

$$\frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ h & 0 & 1 \\ 0 & 4 & 1 \end{vmatrix} = \pm 2$$

$$\Rightarrow 4h = \pm 4$$

$$\Rightarrow h = \pm 1$$

$$\text{Hence sum} = 1 - 1 = 0$$

4. If the points (a, b) , $(1, 1)$ and $(2, 2)$ are collinear then

(1) $a = 2b$

(2) $a = 3b$

(3) $2a = b$

(4) $a = b$

Sol. Answer (4)

$$\begin{vmatrix} a & b & 1 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow a(1-2) - b(1-2) + 1(2-2) = 0$$

$$\Rightarrow -a + b = 0 \Rightarrow a = b$$

5. If the sides of triangle ABC are such that $a = 4$, $b = 5$, $c = 6$, then the ratio in which incentre divide the angle bisector of B is

(1) $2 : 3$

(2) $2 : 1$

(3) $5 : 2$

(4) $1 : 1$

Sol. Answer (2)

$$\text{Ratio} = \frac{c+a}{b} = \frac{4+6}{5} = 2:1$$

6. If the sides of triangle ABC are such that $a = 4$, $b = 5$, $c = 6$ then the ratio in which excentre corresponding to A divides the angle bisector AD in the ratio

(1) $11 : 4$ externally

(2) $4 : 11$ externally

(3) $3 : 2$ externally

(4) $2 : 3$ externally

Sol. Answer (1)

$$\text{Ratio} = \frac{b+c}{a} = 11:4 \text{ externally}$$

7. Let in triangle ABC , $A = 45^\circ$, $B = 60^\circ$, $C = 75^\circ$ then the ratio in which the orthocentre divides the altitude AD is

(1) $(2+2\sqrt{3}) : 1$

(2) $1 : (2+2\sqrt{3})$

(3) $1 : 2$

(4) $2 : 3$

Sol. Answer (1)

$$\text{Ratio} = \frac{\tan B + \tan C}{\tan A} = \frac{\sqrt{3} + 2 + \sqrt{3}}{1} = 2 + 2\sqrt{3}$$

8. Which of the following is not always inside a triangle ?

(1) Incentre

(2) Centroid

(3) Intersection of altitudes

(4) Intersection of medians

Sol. Answer (3)

Orthocentre is not always inside the triangle

9. If the coordinates of vertices of a triangle is always rational then the triangle cannot be

(1) Scalene

(2) Isosceles

(3) Rightangle

(4) Equilateral

Sol. Answer (4)

$$\text{Let } A = (x_1, y_1), B = (x_2, y_2), C = (x_3, y_3)$$

$$A = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \text{Rational number}$$

But the area of equilateral triangle is also calculated by

$$A = \frac{\sqrt{3}}{4} \times (\text{side})^2 = \text{Irrational}$$

Hence triangle cannot be equilateral.

10. If $P(1, 2)$, $Q(4, 6)$, $R(5, 7)$ and $S(a, b)$ are the vertices of a parallelogram $PQRS$ then

- (1) $a = 2, b = 4$ (2) $a = 3, b = 4$ (3) $a = 2, b = 3$ (4) $a = 3, b = 3$

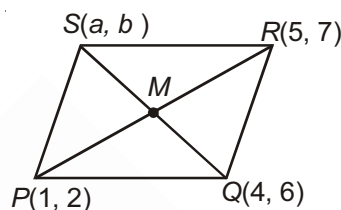
Sol. Answer (3)

In a parallelogram, mid-points coincide

Mid-point of PR = Mid-point of SQ

$$\left(\frac{5+1}{2}, \frac{2+7}{2} \right) = \left(\frac{a+4}{2}, \frac{b+6}{2} \right)$$

$$\Rightarrow a = 2, b = 3$$



11. The area of a triangle whose vertices are (a, a) , $(a + 1, a + 1)$, $(a + 2, a)$ is

- (1) a^2 (2) $2a$ (3) 1 (4) $\sqrt{2}$

Sol. Answer (3)

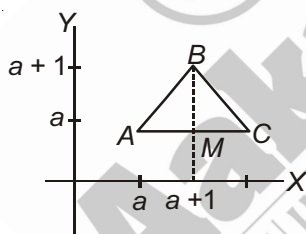
$A(a, a)$

$B(a + 1, a + 1)$

$C(a + 2, a)$

$AC = 2$

$BM = 1$



$$\therefore \text{Area of } \Delta = \frac{1}{2} AC \cdot BM = \frac{1}{2} (2) (1) = 1 \text{ square unit}$$

[Slope, Angle between Lines and Different form of Straight Line]

12. If a line is equally inclined to the axis, then its slope is

- (1) 1 (2) -1 (3) ± 1 (4) 2

Sol. Answer (3)

Slope = ± 1

13. If the angle between two lines is 45° and the slope of one line is 2 , then the product of possible slopes of other line is

- (1) -1 (2) 1 (3) 2 (4) 6

Sol. Answer (1)

$$\tan 45^\circ = \pm \frac{m - 2}{1 + 2m}$$

$$\Rightarrow m = 3, -\frac{1}{3}$$

$$\text{Product} = -1$$

14. If m is the slope of a line and $m + 2 = m + 3$, then the possible angle between line and x -axis is

- (1) 0° (2) 90° (3) 60° (4) 45°

Sol. Answer (2)

In this case the line will be parallel to y -axis. Hence the angle = 90° .

15. The intercept on x -axis of the line $y = mx + c$ is

- (1) c (2) $-c$ (3) $-\frac{c}{m}$ (4) $-\frac{m}{c}$

Sol. Answer (3)

$$0 = mx + c$$

$$\Rightarrow x = -\frac{c}{m}$$

16. The equation of the line having slope = 3 and passing through the point (3, 4) is

- (1) $y + 3x + 4 = 0$ (2) $x + y - 7 = 0$ (3) $y - 3x + 5 = 0$ (4) $x - y + 7 = 0$

Sol. Answer (3)

$$y - 4 = 3(x - 3)$$

$$\Rightarrow y - 3x + 5 = 0$$

17. If $x = x_1 \pm r \cos \theta$, $y = y_1 \pm r \sin \theta$ be the equation of straight line then, the parameter in this equation is

- (1) θ (2) x_1 (3) y_1 (4) r

Sol. Answer (4)

18. The diagonals of parallelogram $PQRS$ are along the lines $x + 3y = 4$, $6x - 2y = 7$, then $PQRS$ must be a

- (1) Rectangle (2) Square (3) Rhombus (4) Trapezium

Sol. Answer (3)

Slopes of diagonals are $-\frac{1}{3}$ and 3.

$$\therefore m_1 m_2 = \left(-\frac{1}{3}\right)(3) = -1 \therefore \text{Diagonals are perpendicular.}$$

$\therefore$ Parallelogram is a Rhombus.

19. What is / are the point(s) on the line $x + y = 4$ that lie(s) at unit distance from the line $4x + 3y = 10$?

- (1) (11, -7) (2) (3, 1) (3) (-7, 11) & (3, 1) (4) (7, 11) & (1, -3)

Sol. Answer (3)

Let point on line $x + y = 4$ be $(x, 4 - x)$

$$\therefore 1 = \frac{|4x + 3(4 - x) - 10|}{\sqrt{16 + 9}}$$

$$|x + 2| = 5 \Rightarrow x = 3 \text{ \& } -7$$

$\therefore$ Points are (3, 1) & (-7, 11)

20. The line $5x + 4y = 0$ passes through the point of intersection of straight lines

(1) $x + 2y - 10 = 0$, $2x + y = -5$

(2) $x + 2y + 10 = 0$, $2x - y + 5 = 0$

(3) $x - 2y - 10 = 0$, $2x + y - 5 = 0$

(4) $x = y$, $2x = y + 1$

Sol. Answer (1)

Equation of a lines which passes through the intersection of two lines

$$L_1 + \lambda L_2 = 0$$

Let $L_1 \equiv x + 2y - 10 = 0$

$$L_2 \equiv 2x + y + 5 = 0$$

$$L_1 + 2L_2 = 0$$

$$(x + 2y - 10) + 2(2x + y + 5) = 0$$

$$5x + 4y = 0$$

21. The vertices of a triangle are $A(-1, -7)$, $B(5, 1)$ and $C(1, 4)$. The equation of angle bisector of $\angle ABC$ is

(1) $x + 7y + 2 = 0$

(2) $x - 7y + 2 = 0$

(3) $x - 7y - 2 = 0$

(4) $x + 7y - 2 = 0$

Sol. Answer (2)

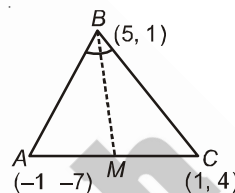
$$\frac{AM}{CM} = \frac{AB}{BC} = \frac{10}{5} = 2$$

$\therefore$ By section formula the coordinates of M

$$\left(\frac{2(-1) + 8 - 7}{2+1}, \frac{8-7}{2+1} \right) = \left(\frac{1}{3}, \frac{1}{3} \right)$$

$$\therefore \text{Equation of } BM = \frac{x-5}{\frac{1}{3}-5} = \frac{y-1}{\frac{1}{3}-1}$$

$$x - 7y + 2 = 0$$



22. In $\triangle ABC$, if $A \equiv (1, 2)$ and equations of the medians through B and C are $x + y = 5$ and $x = 4$, then point B must be

(1) $(1, 4)$

(2) $(7, -2)$

(3) $(4, 1)$

(4) $(-2, 7)$

Sol. Answer (2)

Equation of CN be $x = 4$

$\therefore$ Let coordinate of N be $(4, b)$

N is mid point of AB

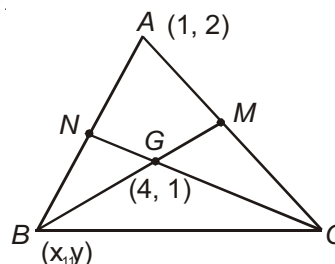
$\therefore$ Coordinate of $B(7, 2b - 2)$

B lie on the line $x + y = 5$

$$\therefore 7 + 2b - 2 = 5$$

$$b = 0$$

$$\therefore B(7, -2)$$



[Length of Perpendicular, Angle Bisectors, Reflection and Analysis of Three Lines]

23. The equation of base of an equilateral triangle is $x + y = 2$ and vertex is $(2, -1)$, then the length of the side of the triangle is equal to

(1) $\sqrt{\frac{2}{3}}$

(2) $\sqrt{\frac{1}{3}}$

(3) $\sqrt{\frac{3}{2}}$

(4) $\sqrt{3}$

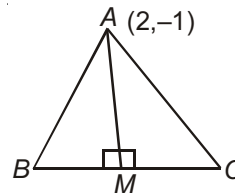
Sol. Answer (1)Equation of BC

$$x + y - 2 = 0$$

$$AM = \left| \frac{2-1-2}{\sqrt{1+1}} \right| = \frac{1}{\sqrt{2}} \text{ [Altitude of equatorial } \Delta = \frac{\sqrt{3}}{2} \text{ side]}$$

$$\frac{\sqrt{3}}{2}(\text{side}) = \frac{1}{\sqrt{2}}$$

$$\text{Side} = \sqrt{\frac{2}{3}}$$



24. The equation of a straight line equally inclined to the axes and equidistance from the points $(1, -2)$ and $(3, 4)$ is

(1) $x + y + 1 = 0$

(2) $x - y + 1 = 0$

(3) $x - y - 1 = 0$

(4) $x + y - 1 = 0$

Sol. Answer (3)Slope of line $= \pm 1$ Let equation of straight line be $y = x + C$

$$\therefore x - y + C = 0 \quad \dots(1)$$

Line (1) is equidistance from points $(1, -2)$ & $(3, 4)$

$$\therefore \left| \frac{1+2+C}{\sqrt{2}} \right| = \left| \frac{3-4+C}{\sqrt{2}} \right|$$

$$|C + 3| = |C - 1|$$

$$C + 3 = -C + 1 \Rightarrow C = -1$$

From equation (1) line is $x - y - 1 = 0$

25. A line passing through $(0, 0)$ and perpendicular to $2x + y + 6 = 0$, $4x + 2y - 9 = 0$ then the origin divides the line in the ratio of

(1) $1 : 2$

(2) $2 : 1$

(3) $4 : 3$

(4) $3 : 4$

Sol. Answer (3)

Perpendicular distances of the lines from origin are

$$OM = \frac{6}{\sqrt{5}} \text{ and } ON = \left| \frac{-9}{\sqrt{20}} \right| = \frac{9}{2\sqrt{5}}$$

$$O \text{ divides } MN \text{ in the ratio} = \frac{6}{\sqrt{5}} : \frac{9}{2\sqrt{5}} = 2 : \frac{3}{2} = 4 : 3$$

26. A ray of light is sent along the line $x - 2y - 3 = 0$ on reaching the line $3x - 2y - 5 = 0$ the ray is reflected from it, then the equation of the line containing the reflected ray is

(1) $2x - 29y - 30 = 0$

(2) $29x - 2y - 31 = 0$

(3) $3x - 31y + 37 = 0$

(4) $31x - 3y + 37 = 0$

Sol. Answer (2)Let A be the point of incidence $\therefore A$ is intersection of

$$x - 2y - 3 = 0 \quad \dots(i)$$

$$\text{and } 3x - 2y - 5 = 0 \quad \dots(ii)$$

$$\therefore A = (1, -1)$$

Let P be any point on the line of incidence $x - 2y - 3 = 0$. So we take $P = (3, 0)$

Let $Q(\alpha, \beta)$ be image of P in the line $3x - 2y - 5 = 0$

$$\therefore PQ \perp \text{the line } 3x - 2y - 5$$

$$\therefore \frac{\beta}{\alpha - 3} \times \frac{B}{2} = -1 \quad \dots(iii)$$

$$\text{And } 3\left(\frac{\alpha + 3}{2}\right) - 2\left(\frac{\beta}{2}\right) - 5 = 0 \quad \dots(iv)$$

$$\text{Equation (iii)} \Rightarrow 3\beta + 2\alpha = 6$$

$$\text{Equation (iv)} \Rightarrow 3\alpha - 2\beta - 1 = 0$$

$$\text{Solving these we get } \alpha = \frac{15}{13}, \beta = \frac{16}{13}$$

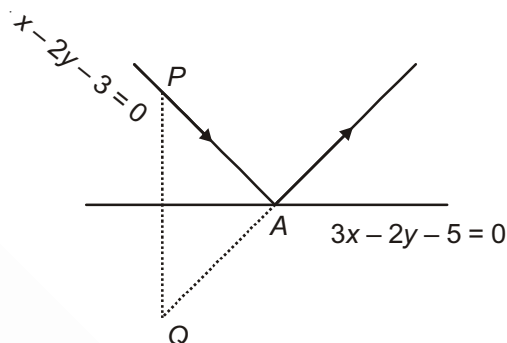
$$\therefore Q = \left(\frac{15}{13}, \frac{16}{13}\right)$$

$$\therefore \text{Line containing the reflected ray is the line joining the points } A(1, -1) \text{ and } Q\left(\frac{15}{13}, \frac{16}{13}\right)$$

$\therefore$ Required equation is

$$y + 1 = \frac{\frac{16}{13} + 1}{\frac{15}{13} - 1} (x - 1)$$

$$\Rightarrow 29x - 2y - 31 = 0$$



27. The equation of the bisector of the acute angle between the lines $3x - 4y + 7 = 0$ and $12x + 5y - 2 = 0$ is

- (1) $11x + 3y - 9 = 0$ (2) $3x - 11y + 9 = 0$ (3) $11x - 3y - 9 = 0$ (4) $11x - 3y + 9 = 0$

Sol. Answer (4)

$$\text{Lines are } 3x - 4y + 7 = 0$$

$$-12x - 5y + 2 = 0$$

$$a_1 a_2 + b_1 b_2 = -36 + 20 < 0$$

$$\text{positive sign gives acute angle bisector, } \frac{3x - 4y + 7}{\sqrt{9 + 16}} = + \frac{-12x - 5y + 2}{\sqrt{144 + 25}}$$

$$11x - 3y + 9 = 0$$

28. Line $ax + by + p = 0$ makes an angle $\frac{\pi}{4}$ with $x \cos \alpha + y \sin \alpha = p$, $p \in \mathbb{R}^+$. If these lines and the line $x \sin \alpha - y \cos \alpha = 0$ are concurrent, then

- (1) $a^2 + b^2 = 1$ (2) $a^2 + b^2 = 2$ (3) $2(a^2 + b^2) = 1$ (4) $a^2 + b^2 = 3$

Sol. Answer (2)

Lines $x\cos\alpha + y\sin\alpha = p$ and $x\sin\alpha - y\cos\alpha = 0$ are perpendicular to each other. Thus $ax + by + p = 0$ is equally inclined to these lines and it will be the angle bisector of these lines. Now equations of angle bisectors is

$$x\sin\alpha - y\cos\alpha = \pm (x\cos\alpha + y\sin\alpha - p)$$

$$\Rightarrow x(\cos\alpha - \sin\alpha) + y(\sin\alpha + \cos\alpha) = p$$

$$\text{or } x(\sin\alpha + \cos\alpha) - y(\cos\alpha - \sin\alpha) = p$$

Comparing these lines with $ax + by + p = 0$, we get

$$\frac{a}{\cos\alpha - \sin\alpha} = \frac{b}{\sin\alpha + \cos\alpha} = 1$$

$$\Rightarrow a^2 + b^2 = 2 \quad \text{or} \quad \frac{a}{\sin\alpha + \cos\alpha} = \frac{b}{\sin\alpha - \cos\alpha} = 1$$

$$\Rightarrow a^2 + b^2 = 2$$

29. The equation of three lines PQ , RS and LM are $(p - q)x + (q - r)y + (r - p) = 0$, $(q - r)x + (r - p)y + p - q = 0$ and $(r - p)x + (p - q)y + (q - r) = 0$ respectively. Which one of the following inferences is correct?

- (1) The lines are parallel to each other
(2) PQ and RS are perpendicular to LM
(3) The lines are concurrent
(4) All the lines are coincident

Sol. Answer (3)

We observe that the sum of coefficients in all equations is zero.

Hence the lines are concurrent at $(1, 1)$.

30. If a , b and c are in A.P. then $ax + by + c = 0$ will always pass through a fixed point whose co-ordinates are

- (1) $(1, -2)$ (2) $(1, 2)$ (3) $(0, -2)$ (4) $(1, -1)$

Sol. Answer (1)

$$b = \frac{a+c}{2}$$

$$\therefore ax + \left(\frac{a+c}{2}\right)y + c = 0$$

$$a\left(x + \frac{1}{2}y\right) + c\left(\frac{1}{2}y + 1\right) = 0$$

$$\left(x + \frac{1}{2}y\right) + \frac{c}{a}\left(\frac{1}{2}y + 1\right) = 0$$

This line will always pass through the intersection point of two lines

$$x + \frac{1}{2}y = 0 \quad \text{and} \quad \frac{1}{2}y + 1 = 0$$

Solve these equations $y = -2$, $x = 1$

Fixed point $(1, -2)$

31. Sum of the possible values of λ for which the following three lines $x + y = 1$, $\lambda x + 2y = 3$, $\lambda^2 x + 4y + 9 = 0$ are concurrent is

- (1) -13 (2) 14 (3) 16 (4) -14

Sol. Answer (1)

The given lines are concurrent if

$$\begin{vmatrix} 1 & 1 & -1 \\ \lambda & 2 & -3 \\ \lambda^2 & 4 & 9 \end{vmatrix} = 0$$

Solving we get

$$\lambda^2 + 13\lambda - 30 = 0$$

Which gives two values of λ whose sum is -13 **[Pair of Straight Lines]**32. The equation $2x^2 + 3y^2 - 8x - 18y + 35 = K$ represents

- (1) No locus if $K > 0$ (2) An ellipse if $K < 0$ (3) A point if $K = 0$ (4) A hyperbola if $K > 0$

Sol. Answer (3)By complete squaring method $2(x - 2)^2 + 3(y - 3)^2 = k$ If $k = 0$

$$2(x - 2)^2 + 3(y - 3)^2 = 0$$

Then necessarily $(x - 2)^2 = 0$ & $(y - 3)^2 = 0$ $\therefore x = 2$ & $y = 3$ $\therefore$ Equation represents a point if $k = 0$ 33. If $x^2 - 2pxy - y^2 = 0$, $x^2 - 2qxy - y^2 = 0$ are bisectors of each other, then

- (1) $p = q + 1$ (2) $pq = -1$ (3) $p + q + 1 = 0$ (4) $pq = 2$

Sol. Answer (2)Equation of angle bisectors of $x^2 - 2pxy - y^2 = 0$

$$\text{will be } \frac{x^2 - y^2}{1 - (-1)} = \frac{xy}{-p}$$

$$px^2 + 2xy - py^2 = 0 \quad \dots(1)$$

$$x^2 - 2qxy - y^2 = 0 \quad \dots(2)$$

Represents coincident lines

$$\frac{p}{1} = \frac{2}{-2q} = \frac{-p}{-1} \quad \therefore pq = -1$$

34. The equation $\sqrt{x^2 + 4y^2 - 4xy + 4} + x - 2y = 1$ represents a/an

- (1) Straight line (2) Ellipse (3) Circle (4) Parabola

Sol. Answer (1)

$$\sqrt{x^2 + 4y^2 - 4xy + 4} = 1 + 2y - x$$

Squaring both sides

$$x^2 + 4y^2 - 4xy + 4 = 1 + 4y^2 + x^2 + 4y - 4xy - 2x$$

$$2x - 4y + 3 = 0$$

Equation represents a straight line

35. The equation of lines joining (0, 0) and points of intersection of $x^2 + y^2 + 2xy = 4$, $3x^2 + 5y^2 - xy = 7$ is
 (1) $x^2 + y^2 - xy = 0$ (2) $5x^2 + 2y^2 - 13xy = 0$ (3) $5x^2 + y^2 - 6xy = 0$ (4) $5x^2 + 13y^2 - 18xy = 0$

Sol. Answer (4)

Equation of pair of lines which passes through the intersection of given curves be $L_1 + \lambda L_2 = 0$

$$(x^2 + y^2 + 2xy - 4) + \lambda (3x^2 + 5y^2 - xy - 7) = 0$$

$$(1 + 3\lambda)x^2 + (1 + 5\lambda)y^2 + (2 - \lambda)xy - (4 + 7\lambda) = 0$$

Lines passes through origin

$\therefore$ Equation should be homogeneous

$$\therefore \text{Put } 4 + 7\lambda = 0$$

$$\Rightarrow \lambda = -\frac{4}{7}$$

$$\therefore \text{Equation of lines is } 5x^2 + 13y^2 - 18xy = 0$$

36. The straight line $ax + by = 1$ makes with the curve $px^2 + 2axy + qy^2 = r$, a chord which subtends a right angle at the origin. Then

$$(1) r(b^2 + q^2) = p + a \quad (2) r(b^2 + p^2) = p + q \quad (3) r(a^2 + b^2) = p + q \quad (4) (a^2 + p^2)r = q + b$$

Sol. Answer (3)

$$px^2 + 2axy + qy^2 = r(1)^2$$

$$px^2 + 2axy + qy^2 = r[ax + by]^2$$

$$(p - ra^2)x^2 + (q - rb^2)y^2 + (a - rab)2xy = 0$$

These lines are perpendicular

$$\therefore p - ra^2 + q - rb^2 = 0$$

$$p + q = r(a^2 + b^2)$$

37. If two of the lines represented by

$$x^4 + x^3y + cx^2y^2 - xy^3 + y^4 = 0$$

$$(1) 0 \quad (2) -1 \quad (3) 1 \quad (4) -6$$

Sol. Answer (4)

Since the product of the slopes of the four lines represented by the given equation is 1 and a pair of lines represents the bisector of the angles between the other two. The product of the slopes of each pair is -1. So let equation of one pair be $ax^2 + 2hxy - ay^2 = 0$

The equation of its bisector is

$$\frac{x^2 - y^2}{2a} = \frac{xy}{h}$$

By hypothesis,

$$x^4 + x^3y + cx^2y^2 - xy^3 + y^4$$

$$= (ax^2 + 2hxy - ay^2) \times (hx^2 - 2axy - hy^2)$$

$$= ah(x^4 + y^4) + 2(h^2 - a^2)(x^3y - xy^3) - 6ahx^2y^2$$

Comparing the respective coefficients, we get

$$ah = 1, c = -6ah = -6$$

38. The changed equation of locus $x^2 + 6xy + y^2 = 1$ when the lines $x + y = 0$ and $x - y + 1 = 0$ are taken as the new x and y axis respectively is given by

$$(1) \quad x^2 - 2y^2 - \sqrt{2}x + 1 = 0$$

$$(2) \quad x^2 - 4y^2 + \sqrt{3}x - 1 = 0$$

$$(3) \quad 2x^2 - y^2 - \sqrt{2}x + 2 = 0$$

$$(4) \quad 2x^2 - 3y^2 - \sqrt{5}y - 1 = 0$$

Sol. Answer (1)

Here the lines $x + y = 0$ and $x - y + 1 = 0$ are perpendicular to each other

$$\text{So take } x' = \frac{x - y + 1}{\sqrt{2}} \text{ and } y' = \frac{x + y}{\sqrt{2}}$$

$$\Rightarrow x - y + 1 = \sqrt{2}x' \quad \dots(i)$$

$$x + y = \sqrt{2}y' \quad \dots(ii)$$

Solving (i) & (ii), we get

$$x = \frac{x' + y'}{\sqrt{2}} - \frac{1}{2} \text{ and } y = \frac{y' - x'}{\sqrt{2}} + \frac{1}{2}$$

Putting these value in the given locus we get

$$x'^2 - 2y'^2 - \sqrt{2}x' + 1 = 0$$

Changing (x', y') into (x, y) we get

$$x^2 - 2y^2 - \sqrt{2}x + 1 = 0$$

39. The equation of a pair of straight lines is $ax^2 + 2hxy + by^2 = 0$. The axes are rotated by angle ϕ so that term containing xy in the equation may be removed, Then ϕ must be equal to

$$(1) \quad \frac{1}{2} \tan^{-1} \frac{3h}{b-a}$$

$$(2) \quad \frac{1}{2} \tan^{-1} \frac{2h}{a-b}$$

$$(3) \quad \frac{1}{3} \tan^{-1} \frac{2h}{a-b}$$

$$(4) \quad \frac{1}{3} \tan^{-1} \frac{3h}{b-a}$$

Sol. Answer (2)

Here the origin remains fixed and axes are rotated through angle ϕ , in anticlockwise sense

Let new co-ordinates of the point (x, y) becomes (x', y')

Then equation of transformation will be

$$x = x' \cos \phi - y' \sin \phi$$

$$y = x' \sin \phi + y' \cos \phi$$

$\therefore$ Changed equation will be

$$a(x' \cos \phi - y' \sin \phi)^2 + 2h(x' \cos \phi - y' \sin \phi)(x' \sin \phi + y' \cos \phi) + b(x' \sin \phi + y' \cos \phi)^2 = 0$$

(Since this expression is free from xy)

$$\therefore -2a \cdot \sin \phi \cdot \cos \phi + 2h(\cos^2 \phi - \sin^2 \phi) + 2b \sin \phi \cdot \cos \phi = 0$$

$$-a \cdot \sin 2\phi + 2h \cdot \cos 2\phi + b \cdot \sin 2\phi = 0$$

$$\Rightarrow \sin 2\phi (a - b) = 2h \cdot \cos 2\phi$$

$$\Rightarrow \tan 2\phi = \frac{2h}{a-b}$$

$$\Rightarrow 2\phi = \tan^{-1} \left(\frac{2h}{a-b} \right)$$

$$\Rightarrow \phi = \frac{1}{2} \tan^{-1} \left(\frac{2h}{a-b} \right)$$

40. If one equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines. then the product of the distances of the lines from origin is

(1) $\frac{2c}{\sqrt{(a-b)^2 + 4h^2}}$ (2) $\frac{2c}{\sqrt{(a-b)^2 + 3h^2}}$ (3) $\frac{c}{\sqrt{(a-b)^2 + 4h^2}}$ (4) $\frac{c}{\sqrt{(a-b)^2 + 3h^2}}$

Sol. Answer (3)

The given equation is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c \equiv (a_1x + b_1y + c_1)(a_2x + b_2y + c_2) = 0$$

Where

$$a_1a_2 = a \quad a_1b_2 + b_1a_2 = 2h$$

$$b_1b_2 = b \quad a_1c_2 + c_1a_2 = 2g$$

$$c_1c_2 = c \quad b_1c_2 + c_1b_2 = 2f$$

Lines are

$$a_1x + b_1y + c_1 = 0 \quad \dots(i)$$

$$\text{And } a_2x + b_2y + c_2 = 0 \quad \dots(ii)$$

Product of distances of lines (i) and (ii) from origin is given by

$$\begin{aligned} &= \frac{c_1}{\sqrt{a_1^2 + b_1^2}} \cdot \frac{c_2}{\sqrt{a_2^2 + b_2^2}} = \frac{c_1 \cdot c_2}{\sqrt{a_1^2 \cdot a_2^2 + a_1^2 \cdot b_2^2 + b_1^2 \cdot a_2^2 + b_1^2 \cdot b_2^2}} \\ &= \frac{c}{\sqrt{a^2 + b^2 + 4h^2 - 2ab}} \\ &= \frac{c}{\sqrt{(a-b)^2 + 4h^2}} \end{aligned}$$

41. The straight lines joining the origin to the point of intersection of two curves

$$ax^2 + 2hxy + by^2 + 2gx = 0$$

and $a'x^2 + 2h'xy + b'y^2 + 2g'x = 0$ will be at right angles to one another if

(1) $g'(a+b) = g(a'+b')$ (2) $g'(a'+b) = g(a+b')$ (3) $g'(a-b) = g(a'-b')$ (4) $g'(a-b') = g(a'-b)$

Sol. Answer (1)

Any curve passing through the intersection of the given curves is

$$ax^2 + 2hxy + by^2 + 2gx + \lambda(a'x^2 + 2h'xy + b'y^2 + 2g'x) = 0 \quad \dots(i)$$

This will be pair of straight lines passing through origin if it is IInd degree homogeneous in x and y. For this the condition on (i) is

$$\text{Coefficient of } x = 2g + 2\lambda g' = 0 \Rightarrow \lambda = \frac{-g}{g'}$$

Also the lines are perpendicular

i.e., coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow a + \lambda a' + b + \lambda b' = 0$$

$$\Rightarrow a + b + \lambda(a' + b') = 0$$

$$\Rightarrow a + b = -\lambda(a' + b')$$

$$\Rightarrow g'(a+b) = g(a'+b')$$

42. The line AB whose equation is $x - y = 2$ cuts the x -axis at A and B is $(4, 2)$. The line segment AB is rotated about 'A' through an angle 45° in the anticlockwise sense. Then the equation AB in new position is

- (1) $x + 2 = 0$ (2) $x - 2 = 0$ (3) $x - \sqrt{2}y - 2 = 0$ (4) $y - 2 = 0$

Sol. Answer (2)

The given equation of the line is $x - y = 2$

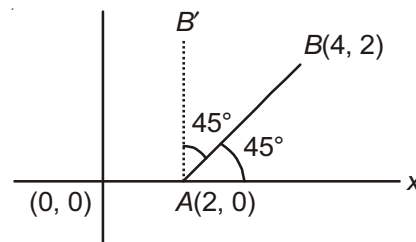
$A = (2, 0)$ and $B = (4, 2)$

$$\angle BAX = \tan^{-1}\left(\frac{2-0}{4-2}\right) = 45^\circ$$

$$\therefore \angle B'AX = \frac{\pi}{2}, \text{ where } B' \text{ is the new position of } B \text{ so } \theta = \frac{\pi}{2}$$

Where AB' makes an angle θ with +ve direction of x -axis

$$\therefore \text{Equation of } AB' = x - 2 = 0$$



43. The middle point of the line segment joining $(3, -1)$ and $(1, 1)$ is shifted by two units (in the sense increasing y) perpendicular to the line segment. Then the coordinates of the point in the new position is

- (1) $(2 - \sqrt{2}, \sqrt{2})$ (2) $(\sqrt{2}, 2 + \sqrt{2})$ (3) $(2 + \sqrt{2}, \sqrt{2})$ (4) $(\sqrt{2}, 2 - \sqrt{2})$

Sol. Answer (3)

Let P be the middle point of the line segment joining $A(3, -1)$ and $B(1, 1)$

Then $P = (2, 0)$

Let P be shifted to Q where $PQ = 2$ and y -coordinate of Q is greater than that of P (from graph)

$$\text{Now, Slope of } AB = \frac{1+1}{1-3} = -1$$

$$\therefore \text{Slope of } PQ = 1$$

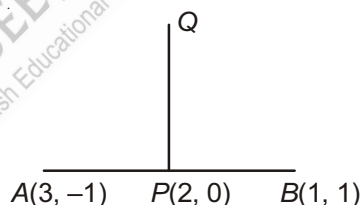
$\therefore$ Coordinates in Q by distance formula

$$= (2 \pm 2\cos\theta, 0 \pm 2\sin\theta), \text{ where } \tan\theta = 1$$

$$= (2 \pm \sqrt{2}, \pm \sqrt{2})$$

As y -coordinate of Q is greater than that of P

$$\therefore Q = (2 + \sqrt{2}, \sqrt{2}), \text{ which is the required point.}$$



44. Distance between the pair of straight lines

$x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ is given by

- (1) $\frac{3}{\sqrt{10}}$ (2) $\frac{5}{\sqrt{10}}$ (3) $\frac{6}{\sqrt{10}}$ (4) $\frac{7}{\sqrt{10}}$

Sol. Answer (3)

The given equation is

$$x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0 \quad \dots(i)$$

$$\text{Here } abc + 2gfh - af^2 - bg^2 - ch^2 = 0$$

And $h^2 - ab = 0$

$\therefore$ Equation (i) Represents the parallel straight lines

From (i) we know

$$9y^2 + 6(x+2)y + (x^2 + 4x - 5) = 0$$

$$y = \frac{-6(x+2) \pm \sqrt{36(x+2)^2 - 36(x^2 + 4x - 5)}}{2 \times 9}$$

$$\text{Or } 3y + x = 1 \text{ and } 3y + x + 5 = 0$$

There are two parallel lines and distance between these two lines is

$$\frac{5 - (-1)}{\sqrt{3^2 + 1^2}} = \frac{6}{\sqrt{10}}$$

[Miscellaneous]

45. $y = 10^x$ is the reflection of $y = \log_{10} x$ in the line whose equation is

- (1) $y = x$ (2) $y = -x$ (3) $y = 10^x$ (4) $y = -10x$

Sol. Answer (1)

Reflection of $y = \log_{10} x$

About $y = x$ is $x = \log_{10} y$

$$\Rightarrow y = 10^x$$

46. The limiting position of the point of intersection of the straight line $3x + 5y = 1$ and $(2+c)x + 5c^2y = 1$ as c tends to one is

- (1) $\left(\frac{1}{2}, -\frac{1}{10}\right)$ (2) $\left(\frac{3}{8}, -\frac{1}{40}\right)$ (3) $\left(\frac{2}{5}, \frac{1}{25}\right)$ (4) $\left(\frac{2}{5}, -\frac{1}{25}\right)$

Sol. Answer (4)

$$\text{Solve these equations } x = \frac{(C^2 - 1)}{(3C + 2)(C - 1)} \text{ and } y = \frac{(C - 1)}{-5(3C + 2)(C - 1)}$$

Here $C \neq 1$ at $C = 1$ lines are coincident.

$$\therefore x = \lim_{C \rightarrow 1} \frac{(C^2 - 1)}{(3C + 2)(C - 1)} = \frac{C + 1}{3C + 2} \Rightarrow x = \frac{2}{5}$$

$$\therefore y = \lim_{C \rightarrow 1} \frac{(C - 1)}{-5(3C + 2)(C - 1)} = \lim_{C \rightarrow 1} \frac{1}{-5(3C + 2)} = -\frac{1}{25}$$

$$\therefore \text{Point of intersection is } \left(\frac{2}{5}, -\frac{1}{25}\right)$$

47. The ends A, B of a straight line segment of constant length c slides on the fixed rectangular axes OX, OY respectively. If the rectangle $OAPB$ be completed. Then the locus of the foot of the perpendicular drawn from P upon AB is

- (1) $x^{2/3} - y^{2/3} = c^{2/3}$ (2) $x^{1/3} + y^{1/3} = c^{1/3}$ (3) $x^{2/3} + y^{2/3} = c^{2/3}$ (4) $x^{1/3} - y^{2/3} = c^{1/3}$

Sol. Answer (3)

Let $\angle BAO = \theta$, then

$$OA = c \cos \theta$$

$$OB = c \sin \theta$$

Let $m(h, k)$ be foot of the perpendicular from P on AB

Let $MN \perp OX$

$$ON = h = OA - NA$$

$$= c \cos \theta - MA \cdot \cos \theta$$

$$= c \cos \theta - PA \cdot \cos \left(\frac{\pi}{2} - \theta \right) \cdot \cos \theta$$

$$= c \cos \theta - c \sin \theta \cdot \sin \theta \cdot \cos \theta$$

$$= c \cos \theta (1 - \sin^2 \theta)$$

$$\Rightarrow h = c \cos^3 \theta \quad \dots(i)$$

$$k = MN = MA \sin \theta$$

$$\Rightarrow k = c \cdot \sin^3 \theta \quad \dots(ii)$$

$$\Rightarrow h^{2/3} + k^{2/3} = c^{2/3}(\sin^2 \theta + \cos^2 \theta) = c^{2/3}$$

Replacing (h, k) by (x, y) we get

$$x^{2/3} + y^{2/3} = c^{2/3}$$

48. Let a given line $L_1 : ax + by + c = 0$ intersects x -axis and y -axis at P and Q respectively. Let another line L_2 perpendicular to L_1 cut the axes at R and S respectively. The locus of the point of intersection of the lines PS and QR is

$$(1) \quad x^2 + y^2 + \frac{c}{a}x + \frac{b}{a}y = 0$$

$$(2) \quad x^2 + y^2 + \frac{c}{a}x + \frac{c}{b}y = 0$$

$$(3) \quad x^2 + y^2 + \frac{a}{c}x + \frac{b}{c}y = 0$$

$$(4) \quad x^2 + y^2 + \frac{a}{b}x + \frac{c}{b}y = 0$$

Sol. Answer (2)

The given line $L_1 : ax + by + c = 0$

$$\therefore P = \left(\frac{-c}{a}, 0 \right), Q = \left(0, \frac{-c}{b} \right)$$

Any line L_2 is perpendicular to L_1 is

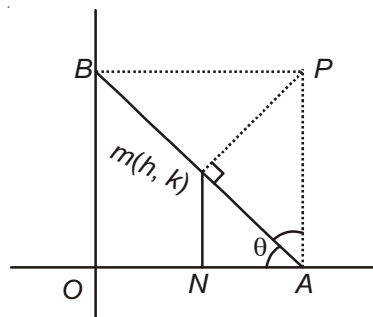
$$bx - ay + \lambda = 0$$

$$R = \left(-\frac{\lambda}{b}, 0 \right), S = \left(0, \frac{\lambda}{a} \right)$$

Equation of line PS is

$$y = \frac{-\lambda/a}{-c/a} \left(x + \frac{c}{a} \right)$$

$$y = \frac{\lambda}{c} \left(x + \frac{c}{a} \right) \quad \dots(i)$$



Equation of line QR is

$$y = \frac{c/b}{-\lambda/b} \left(x + \frac{\lambda}{b} \right)$$

$$y = -\frac{c}{\lambda} \left(x + \frac{\lambda}{b} \right) \quad \dots(ii)$$

Locus of the point of intersection of (i) and (ii) is obtained by eliminating λ from (i) and (ii)

$$\text{From (ii) } y + \frac{c}{b} = \frac{c}{\lambda} x \quad \dots(iii)$$

Multiplying (i) and (iii) we get

$$y \left(y + \frac{c}{b} \right) = x \left(x + \frac{c}{a} \right)$$

$$x^2 + y^2 + \frac{c}{a}x + \frac{c}{b}y = 0$$

49. Vertices of a triangle are $A(x_1, x_1 \tan \alpha_1)$, $B(x_2, x_2 \tan \alpha_2)$ and $C(x_3, x_3 \tan \alpha_3)$. If the circumcentre coincides with the origin and the orthocentre $H = (\bar{x}, \bar{y})$, Then which of the following is necessarily true.

$$(1) \bar{y} (\cos \alpha_1 + \cos \alpha_2 + \cos \alpha_3) = \bar{x} (\sin \alpha_1 + \sin \alpha_2 + \sin \alpha_3)$$

$$(2) \bar{x} (\cos \alpha_1 + \cos \alpha_2 + \cos \alpha_3) = \bar{y} (\sin \alpha_1 + \sin \alpha_2 + \sin \alpha_3)$$

$$(3) \bar{x} (\tan \alpha_1 + \tan \alpha_2 + \tan \alpha_3) = \bar{y} (\cot \alpha_1 + \cot \alpha_2 + \cot \alpha_3)$$

$$(4) \bar{y} (\tan \alpha_1 + \tan \alpha_2 + \tan \alpha_3) = \bar{x} (\tan \alpha_1 + \tan \alpha_2 + \tan \alpha_3)$$

Sol. Answer (1)

Here circumcentre $O = (0, 0)$

So, $OA = OB = OC$

$$\therefore x_1^2 + x_1^2 \tan^2 \alpha_1 = x_2^2 + x_2^2 \tan^2 \alpha_2$$

$$= x_3^2 + x_3^2 \tan^2 \alpha_3$$

$$\Rightarrow x_1^2 \cdot \sec^2 \alpha_1 = x_2^2 \cdot \sec^2 \alpha_2 = x_3^2 \cdot \sec^2 \alpha_3$$

$$\Rightarrow \frac{x_1}{\cos \alpha_1} = \frac{x_2}{\cos \alpha_2} = \frac{x_3}{\cos \alpha_3} = k \text{ (suppose)}$$

$\therefore$ Vertices of the triangle become

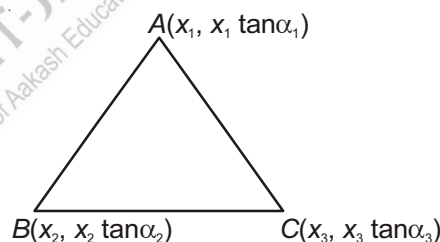
$$A = (k \cos \alpha_1, k \sin \alpha_1)$$

$$B = (k \cos \alpha_2, k \sin \alpha_2)$$

$$C = (k \cos \alpha_3, k \sin \alpha_3)$$

$$\therefore \text{Centroid } G = \left(k \cdot \frac{(\cos \alpha_1 + \cos \alpha_2 + \cos \alpha_3)}{3}, k \cdot \frac{(\sin \alpha_1 + \sin \alpha_2 + \sin \alpha_3)}{3} \right)$$

We know that orthocentre H , centroid G and circumcentre O are collinear



So, Slope of HO = slope of GO

$$\frac{\bar{y}}{\bar{x}} = \frac{\sin \alpha_1 + \sin \alpha_2 + \sin \alpha_3}{\cos \alpha_1 + \cos \alpha_2 + \cos \alpha_3}$$

$$\bar{y}(\cos \alpha_1 + \cos \alpha_2 + \cos \alpha_3) = \bar{x}(\sin \alpha_1 + \sin \alpha_2 + \sin \alpha_3)$$

50. Let $O(0, 0)$, $P(3, 4)$, $Q(6, 0)$ be the vertices of the triangle OPQ . The point R inside the triangle OPQ is such that the triangles OPR , PQR , OQR are of equal area. The coordinates of R are [IIT-JEE 2007]

- (1) $\left(\frac{4}{3}, 3\right)$ (2) $\left(3, \frac{2}{3}\right)$ (3) $\left(3, \frac{4}{3}\right)$ (4) $\left(\frac{4}{3}, \frac{2}{3}\right)$

Sol. Answer (3)

The given point is the centroid of the triangle.

51. Let $ABCD$ be a quadrilateral with area 18, with side AB parallel to the side CD and $AB = 2 CD$. Let AD be perpendicular to AB and CD . If a circle is drawn inside the quadrilateral $ABCD$ touching all the sides, then its radius is [IIT-JEE 2007]

- (1) 3 (2) 2 (3) $\frac{3}{2}$ (4) 1

Sol. Answer (2)

Let $\angle ABC = 2\theta$, and r the radius of the inscribed circle then AB and CD can be expressed in terms of r and θ . Area of quadrilateral that is trapezium in our case, can be get in term of r and θ and then we can solve the equation for r .

52. Consider three points

$$P = (-\sin(\beta - \alpha), -\cos \beta), Q = (\cos(\beta - \alpha), \sin \beta) \text{ and } R = (\cos(\beta - \alpha + \theta), \sin(\beta - \theta)),$$

where $0 < \alpha, \beta, \theta < \frac{\pi}{4}$. Then,

[IIT-JEE 2008]

- (1) P lies on the line segment RQ (2) Q lies on the line segment PR
(3) R lies on the line segment QP (4) P, Q, R are non-collinear

Sol. Answer (4)

For collinear points

$$\Delta = \begin{vmatrix} -\sin(\beta - \alpha) & -\cos \beta & 1 \\ \cos(\beta - \alpha) & \sin \beta & 1 \\ \cos(\beta - \alpha + \theta) & \sin(\beta - \theta) & 1 \end{vmatrix}$$

Clearly $\Delta \neq 0$ for any value of α, β, θ , hence points are non-collinear.

2nd method : (by observation)

$$\begin{array}{c} \xrightarrow{\cos \theta} \quad \xrightarrow{\sin \theta} \\ P \quad \quad R \quad \quad Q \\ (\sin(\beta - \alpha), -\cos \beta) \quad \quad (\cos(\beta - \alpha), \sin \beta) \end{array}$$

$$\text{In the case } R = \left(\frac{\cos \theta \cdot \cos(\beta - \alpha) - \sin \theta \sin(\beta - \alpha)}{\sin \theta + \cos \theta}, \frac{\cos \theta \sin \beta - \sin \theta \cos \beta}{\sin \theta + \cos \theta} \right)$$

$$R \equiv \left(\frac{\cos(\beta - \alpha + \theta)}{\sin\theta + \cos\theta}, \frac{\sin(\beta - \theta)}{\sin\theta + \cos\theta} \right) = (\cos(\beta - \alpha + \theta), \sin(\beta - \theta)), \text{ if } \sin\theta + \cos\theta = 1$$

Which is not possible if $0 < \theta < \frac{\pi}{4}$

Hence points are non-collinear.

53. The locus of the orthocentre of the triangle formed by the lines

$$(1 + p)x - py + p(1 + p) = 0,$$

$$(1 + q)x - qy + q(1 + q) = 0,$$

and $y = 0$, where $p \neq q$, is

[IIT-JEE 2009]

(1) A hyperbola

(2) A parabola

(3) An ellipse

(4) A straight line

Sol. Answer (4)

We have

$$(1 + p)x - py + p(1 + p) = 0$$

$$\Rightarrow \frac{x}{-p} + \frac{y}{1+p} = 1$$

Also,

$$(1 + q)x - qy + q(1 + q) = 0$$

$$\Rightarrow \frac{x}{-q} + \frac{y}{1+q} = 1$$

Equation of AM is

$$y = \frac{-p}{1+p}(x+a)$$

and equation of BM is $y = \frac{-q}{1+q}(x+p)$

$$\Rightarrow \frac{p}{1+p}(x+a) = \frac{q}{1+q}(x+p)$$

$$\Rightarrow x = pq$$

...(i)

$$\text{Also } y = \frac{-p}{1+p}(pq+q)$$

$$\Rightarrow y = -pq$$

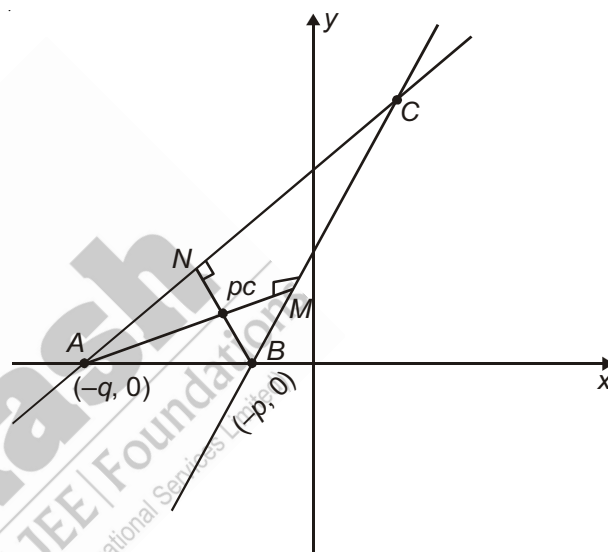
...(ii)

from (i) and (ii)

$$x = -y$$

$$\Rightarrow x + y = 0$$

Which is the required locus representing a straight line.



54. A straight line L through the point $(3, -2)$ is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$. If L also intersects the x -axis, then the equation of L is

[IIT-JEE 2011]

(1) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$

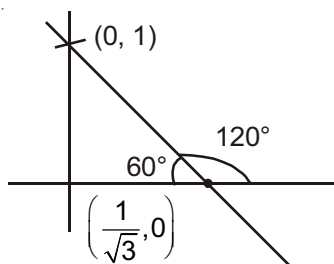
(2) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

(3) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$

(4) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

Sol. Answer (2)

The question is too simple from the diagram, the given line



$\sqrt{3}x + y = 1$ makes an angle 120° with x -axis and intersects at $\left(\frac{1}{\sqrt{3}}, 0\right)$. A line making an angle 60° with the given line is either x -axis or different from x -axis. By observation it is clear that the straight line $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$ is the required line.

Second Solution

The equation of the line through $(-3, -2)$ may be written as

$$y + 2 = m(x + 3)$$

which will make 60° with $\sqrt{3}x + y = 1$ if

$$\Rightarrow \tan 60^\circ = \left| \frac{m + \sqrt{3}}{1 - \sqrt{3}m} \right|$$

$$\Rightarrow \sqrt{3} = \pm \frac{m + \sqrt{3}}{1 - \sqrt{3}m}$$

$$\Rightarrow m = \sqrt{3} \text{ or } m = 0$$

Since the line intersects x -axis also, hence $m \neq 0$ consequently $m = \sqrt{3}$ and the required line is

$$y + 2 = \sqrt{3}(x + 3)$$

$$\Rightarrow y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$$

55. For $a > b > c > 0$, the distance between $(1, 1)$ and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then

[JEE(Advanced) 2013]

(1) $a + b - c > 0$

(2) $a - b + c < 0$

(3) $a - b + c > 0$

(4) $a + b - c < 0$

Sol. Answer (1)

$$ax + by + c = 0 \quad \dots (1)$$

$$bx + ay + c = 0 \quad \dots (2)$$

$$\text{Solving, } x = \frac{-c}{a+b}$$

Also from (1) & (2)

$$y = x$$

∴ Point of intersection lies on $y = x$

$$\Rightarrow y = \frac{-c}{a+b}$$

$$\text{Given, } \sqrt{\left(1 + \frac{c}{a+b}\right)^2 + \left(1 + \frac{c}{a+b}\right)^2} < 2\sqrt{2}$$

$$\Rightarrow \sqrt{2} \left(1 + \frac{c}{a+b}\right) < 2\sqrt{2}$$

$$\Rightarrow \frac{a+b+c}{a+b} < 2$$

$$\Rightarrow a+b+c < 2a+2b$$

$$\Rightarrow a+b-c > 0$$

SECTION - B

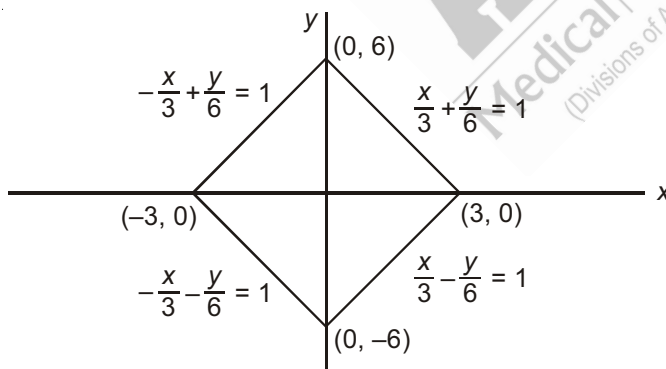
Objective Type Questions (More than one options are correct)

1. If for a line the length of intercept on x-axis is 3 and on y-axis is 6. Then which of the followings is/are true.

- (1) There are four such lines are possible
- (2) The equation of lines is $\pm \frac{x}{3} \pm \frac{y}{6} = 1$
- (3) The figure bounded by these four lines is a rhombus
- (4) The area of figure bounded by all lines is 36

Sol. Answer (1, 2, 3, 4)

$$\text{Area} = 4 \times \frac{1}{2} \times 3 \times 6 = 36$$



Hence all options are correct.

2. Consider triangle ABC where $A = (4, 4)$, $B = (7, 4)$, $C = (4, 7)$ then

- (1) The centroid of triangle ABC is $(5, 5)$
- (2) The orthocentre of triangle ABC is $(4, 4)$
- (3) The circum centre is $\left(\frac{11}{2}, \frac{11}{2}\right)$
- (4) The incentre is $\left(\frac{11+4\sqrt{2}}{2+\sqrt{2}}, \frac{11+4\sqrt{2}}{2+\sqrt{2}}\right)$

Sol. Answer (1, 2, 3, 4)

$$AB = 3$$

$$BC = 3\sqrt{2}$$

$$AC = 3$$

$$\Rightarrow AB^2 + AC^2 = BC^2$$

$$\text{Hence, } \angle A = 90^\circ$$

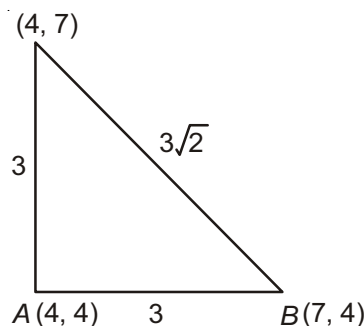
$$\text{Orthocentre} = A = (4, 4)$$

$$\text{Circumcentre} = \left(\frac{11}{2}, \frac{11}{2} \right)$$

$$\text{Centroid} = \left(\frac{4+4+7}{3}, \frac{4+4+7}{3} \right) = (5, 5)$$

$$\text{Incentre} = \left(\frac{3 \times 7 + 3 \times 4 + 3\sqrt{2} \times 4}{3 + 3 + 3\sqrt{2}}, \frac{3 \times 4 + 3\sqrt{2} \times 4 + 4 \times 7}{3 + 3 + 3\sqrt{2}} \right)$$

$$= \left(\frac{11 + 4\sqrt{2}}{2 + \sqrt{2}}, \frac{11 + 4\sqrt{2}}{2 + \sqrt{2}} \right)$$



3. The equation of the lines passes through (2, 3) and making an angle of 45° with the line $2x - y + 3 = 0$ is

(1) $y + 3x = 9$

(2) $3y + 9x = 27$

(3) $3y - x = 7$

(4) $x + y - 5 = 0$

Sol. Answer (1, 2, 3)

Let the slope of the line is m .

$$\tan 45^\circ = \pm \frac{m - 2}{1 + 2m} \Rightarrow m = -3, m = \frac{1}{3}$$

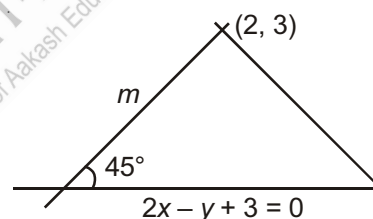
$$\text{Lines } y - 3 = -3(x - 2)$$

$$\boxed{y + 3x = 9}$$

$$\text{and } y - 3 = \frac{1}{3}(x - 2)$$

$$3y - 9 = x - 2$$

$$\boxed{3y - x = 7}$$



4. Consider a point $A(3, 4)$ and line $L \equiv x + 3y + 5 = 0$ then

(1) The image of the point A through the line L is (1, 8)

(2) The perpendicular distance of point A from the line L is $2\sqrt{10}$

(3) Point $(-1, -1)$ lies in the same side as of (3, 4)

(4) The equation of the line which is perpendicular to the line L and passes through the point A is $3x - y - 5 = 0$

Sol. Answer (2, 3, 4)

(A) Let the image is (α, β) then

$$\frac{\alpha - 3}{1} = \frac{\beta - 4}{3} = \frac{-2(3 + 3 \times 4 + 5)}{1 + 9} = \frac{-40}{10} = -\frac{20}{5} = -4$$

$$\alpha = -1, \beta = -8$$

(B) Distance = $\frac{3 + 4 \times 3 + 5}{\sqrt{1 + 9}} = \frac{20}{\sqrt{10}} = 2\sqrt{10}$

(C) True

(D) Equation is $3x - y + k = 0$

At $x = 3, y = 4$

$$9 - 4 + k = 0$$

$$\Rightarrow \boxed{k = -5}$$

Hence equation is $3x - y - 5 = 0$

5. Which of the following is true ?

(1) The equation of the line parallel to $x + y + 2 = 0$ is $x + y + \lambda = 0, \lambda \in R$

(2) The equation of the line perpendicular to $3x + 4y + 7 = 0$ and which passes through $(1, 1)$ is $3y - 4x - 1 = 0$

(3) The distance between the lines $3x + 4y + 5 = 0$ and $6x + 8y + 20 = 0$ is 1

(4) If two lines are parallel and their slopes are real numbers, then their slopes are equal

Sol. Answer (1, 3, 4)

6. If (a, a) lies between the lines $|x + y| = 2$, then a lies in the interval

(1) $(-1, 0]$

(2) $[0, 1)$

(3) $(1, 2)$

(4) $(-2, -1)$

Sol. Answer (1, 2)

$$(a + a - 2)(a + a + 2) < 0 \Rightarrow a \in (-1, 1)$$

7. If the vertices of ΔPQR are rational points, which of the following points of ΔPQR are always rational points?

(1) Centroid

(2) Orthocentre

(3) Circumcenter

(4) Incentre

Sol. Answer (1, 2, 3)

Vertices are rational parts

$$\therefore \text{Centroid} \left(\frac{\sum x_i}{3}, \frac{\sum y_i}{3} \right) \text{ is rational.}$$

Vertices are rational

$\therefore$ Coefficient of equations of lines perpendicular to the sides are also rational

Orthocenter is intersection point of equations of altitudes.

$\therefore$ Orthocenter is rational. Orthocenter, centroid and circumcentre are collinear and centroid divides the line segment in the ratio 2 : 1

$\therefore$ Circumcentre are also rational

$$\text{Incentre} = \left(\frac{px_1 + qx_2 + rx_3}{p + q + r}, \frac{py_1 + qy_2 + ry_3}{p + q + r} \right)$$

Here p, q, r may be irrational

Hence incentre is not always rational.

8. If the line $3x + y + 2 = 0$, $2x - y + 3 = 0$ and $a^2x + 2ay + 6 = 0$ are concurrent then a equals

(1) $\sqrt{7} + 1$

(2) $\sqrt{7} - 1$

(3) $1 - \sqrt{7}$

(4) $a \in R$

Sol. Answer (1, 3)

$$\begin{vmatrix} 3 & 1 & 2 \\ 2 & -1 & 3 \\ a^2 & 2a & 6 \end{vmatrix} = 0$$

$$3 \begin{vmatrix} -1 & 3 \\ 2a & 6 \end{vmatrix} - 1 \begin{vmatrix} 2 & 3 \\ a^2 & 6 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ a^2 & 2a \end{vmatrix} = 0$$

$$a^2 - 2a - 6 = 0$$

$$\Rightarrow a = 1 \pm \sqrt{7}$$

9. A line parallel to the straight line $3x - 4y - 2 = 0$ and at a distance of 4 units from it is

(1) $3x - 4y + 20 = 0$

(2) $4x - 3y + 12 = 0$

(3) $3x - 4y + 18 = 0$

(4) $3x - 4y - 22 = 0$

Sol. Answer (3, 4)

A line parallel to given line

$$3x - 4y + \lambda = 0 \quad \dots(i)$$

$$3x - 4y - 2 = 0 \quad \dots(ii)$$

$$4 = \frac{|\lambda - (-2)|}{\sqrt{9 + 16}}$$

$$|\lambda + 2| = 20 \Rightarrow \lambda = 18, -22$$

$$\therefore \text{Lines are } 3x - 4y + 18 = 0 \text{ \& } 3x - 4y - 22 = 0$$

10. If the point $P(p, q)$ is equidistant from the points $A(a + b, b - a)$ and $B(a - b, a + b)$, then

(1) $ap = bq$

(2) $aq = bp$

(3) $p^2 - q^2 = 2(ap + bq)$

(4) P can be (a, b)

Sol. Answer (2, 4)

$$PA^2 = BP^2$$

$$[p - (a + b)]^2 + [q - (b - a)]^2 = [p - (a - b)]^2 + [q - (a + b)]^2$$

$$aq = bp$$

$$\therefore P(p, q) \text{ can be } (a, b)$$

11. The equation of a bisector of the angle between the lines $y - q = \frac{2a}{1 - a^2}(x - p)$ and $y - q = \frac{2b}{1 - b^2}(x - p)$ is

(1) $(y - q)(a + b) + (x - p)(1 - ab) = 0$

(2) $(y - q)(1 - ab) + (x - p)(a + b) = 0$

(3) $(x - p)(a + b) - (y - q)(1 - ab) = 0$

(4) $(x - p)(a + b) + (y - q)(1 - ab) = 0$

Sol. Answer (1, 3)

Let line (1) makes angle θ_1 , with positive x axis

$$\therefore \tan \theta_1 = \frac{2a}{1 - a^2}$$

$$\theta_1 = \tan^{-1} \left(\frac{2a}{1-a^2} \right) = 2 \tan^{-1} a$$

Let line (ii) makes angle θ_2 with x-axis

$$\therefore \tan \theta_2 = \frac{2b}{1-b^2}$$

$$\therefore \theta_2 = \tan^{-1} \frac{2b}{1-b^2} = 2 \tan^{-1} b$$

$\therefore$ Angle bisector between these two lines makes angle $\theta = \frac{\theta_1 + \theta_2}{2}$ with positive axis.

$$\therefore \theta = \frac{1}{2} \theta_1 + \frac{1}{2} \theta_2 = \tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right) \therefore \tan \theta = \left(\frac{a+b}{1-ab} \right)$$

$\therefore$ Equation of bisector is

$$y - q = \frac{a+b}{1-ab} (x - p)$$

$$(a+b)(x-p) - (1-ab)(y-q) = 0$$

Two bisector are perpendicular

$$\therefore \text{Second bisector is } y - q = \frac{-(1-ab)}{(a+b)} (x - p)$$

$$(1-ab)(x-p) + (a+b)(y-q) = 0$$

12. The point of intersection of the lines $\frac{x}{p} + \frac{y}{q} = 1$ and $\frac{x}{q} + \frac{y}{p} = 1$ lies on the line

$$(1) \ x - y = 0$$

$$(2) \ (x+y)(p+q) = 2pq$$

$$(3) \ (ax+by)(p+q) = (a+b)pq$$

$$(4) \ (ax-by)(p+q) = (a-b)pq$$

Sol. Answer (1, 2, 3, 4)

Point of intersection of lines is $\left(\frac{pq}{p+q}, \frac{pq}{p+q} \right)$ which will satisfy all the four lines.

13. ABC is an isosceles triangle whose base is BC . If B and C are $(a+b, b-a)$ and $(a-b, a+b)$. Then co-ordinates of A may be

$$(1) \ (a, b)$$

$$(2) \ \left(\frac{a}{b}, 1 \right)$$

$$(3) \ \left(\frac{a}{b}, \frac{b}{a} \right)$$

$$(4) \ \left(1, \frac{b}{a} \right)$$

Sol. Answer (2, 4)

$$B(a+b, b-a), C(a-b, a+b)$$

Let M be mid point of BC

$$M(a, b), \text{ Slope of } BC = \frac{2a}{-2b} = -\frac{a}{b}$$

$$\therefore \text{Slope of } AM = \frac{b}{a}$$

Equation of AM, $y - b = \frac{b}{a}(x - a)$

$$\frac{y}{b} - 1 = \frac{x}{a} - 1 \Rightarrow y = \frac{b}{a}x$$

$\left(\frac{a}{b}, 1\right)$, (a, b) and $\left(1, \frac{b}{a}\right)$ will satisfy this equation, but (a, b) is the mid-point of BC. Therefore only $\left(1, \frac{b}{a}\right)$

and $\left(\frac{a}{b}, 1\right)$ can be the required vertex

14. If two vertices of an equilateral triangle are $(1, 1)$ and $(-1, -1)$ then the third vertex may be

(1) $(-\sqrt{3}, -\sqrt{3})$ (2) $(-\sqrt{3}, \sqrt{3})$ (3) $(\sqrt{3}, -\sqrt{3})$ (4) $(\sqrt{3}, \sqrt{3})$

Sol. Answer (2, 3)

Third point of equilateral Δ be

$$\left[\frac{x_1 + x_2 \pm \sqrt{3}(y_1 - y_2)}{2}, \frac{x_1 + x_2 \mp \sqrt{3}(y_1 - y_2)}{2} \right]$$

$$\left[\frac{0 \pm \sqrt{3}(2)}{2}, \frac{0 \mp \sqrt{3}(2)}{2} \right] = (\pm\sqrt{3}, \mp\sqrt{3})$$

15. If the lines $x + 2y + 3 = 0$, $x + 2y - 7 = 0$ and $2x - y - 4 = 0$ form three sides of a square, the equation of fourth side may be

(1) $2x - y - 6 = 0$ (2) $2x - y + 6 = 0$ (3) $2x - y - 14 = 0$ (4) $2x - y + 14 = 0$

Sol. Answer (2, 3)

$x + 2y + 3 = 0$ & $x + 2y - 7 = 0$ are parallel lines

Equation of a line parallel to $2x - y - 4 = 0$ is $2x - y + \lambda = 0$

In a square distance between two parallel lines are equal

$$\therefore \left| \frac{3+7}{\sqrt{1+4}} \right| = \left| \frac{\lambda+4}{\sqrt{4+1}} \right|$$

$$\Rightarrow \lambda + 4 = |10|$$

$$\lambda = 6, -14$$

Lines are $2x - y + 6 = 0$ & $2x - y - 14 = 0$

16. Let L_1 be a straight line passing through $(0, 0)$ and L_2 be $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 and L_2 are equal, then which of the following equations can represent L_1 ?

(1) $x + y = 0$ (2) $x - y = 0$ (3) $x + 7y = 0$ (4) $x + \sqrt{3}y = 0$

Sol. Answer (2, 3)

Centre of circle be $\left(\frac{1}{2}, -\frac{3}{2}\right)$

Let line L_1 be $y = mx$

Intercepts are equal

∴ Lines are at equal distance from the centre

$$\left| \frac{\frac{1}{2}m + \frac{3}{2}}{\sqrt{m^2 + 1}} \right| = \left| \frac{\frac{1}{2} - \frac{3}{2} - 1}{\sqrt{2}} \right|$$

$$|(m + 3)| = 2\sqrt{2} \sqrt{m^2 + 1}$$

Squaring both sides

$$7m^2 - 6m - 1 = 0$$

$$\therefore m = 1, -\frac{1}{7} \quad \therefore \text{Lines are } y = x \text{ \& } y = -\frac{1}{7}x$$

$$x - y = 0 \text{ \& } x + 7y = 0$$

17. Equations of the straight lines inclined at 30° to the x-axis such that the length of its line segments between the co-ordinate axes is 10 units, is

$$(1) x - \sqrt{3}y + 5\sqrt{3} = 0 \quad (2) x + \sqrt{3}y + 5\sqrt{3} = 0 \quad (3) x - \sqrt{3}y - 5\sqrt{3} = 0 \quad (4) x + \sqrt{3}y - 5\sqrt{3} = 0$$

Sol. Answer (1, 3)

$$\text{Slope of line } m = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\therefore \text{Equation of line } y = \frac{1}{\sqrt{3}}x + C$$

Intersection parts on axis are $A(-\sqrt{3}C, 0)$ and $B(0, C)$

Given $AB = 10$

$$\sqrt{3C^2 + C^2} = 10 \quad \Rightarrow C = \pm 5 \quad \therefore \text{Lines are } y = \frac{1}{\sqrt{3}}x \pm 5$$

$$x - \sqrt{3}y \pm 5\sqrt{3} = 0$$

18. $A(1, 3)$ and $C(7, 5)$ are two opposite vertices of a square. The equation of a side through A is

$$(1) x + 2y - 7 = 0 \quad (2) 2x + y - 5 = 0 \quad (3) x - 2y + 5 = 0 \quad (4) 2x - y + 1 = 0$$

Sol. Answer (1, 4)

$$\text{Slope of } AC, m = \frac{5-3}{7-1} = \frac{1}{3}$$

Sides through $A(1, 3)$ makes angle $\frac{\pi}{4}$ with line AC .

∴ Equation of sides are

$$y - 3 = \frac{m \pm \tan \frac{\pi}{4}}{1 \mp m \tan \frac{\pi}{4}} (x - 1)$$

$$y - 3 = \frac{\frac{1}{3} \pm 1}{1 \mp \frac{1}{3}} (x - 1)$$

$$y - 3 = \frac{(1 \pm 3)}{(3 \mp 1)} (x - 1)$$

$\therefore$ Lines are $2x - y + 1 = 0$ & $x + 2y - 7 = 0$

19. The locus represented by the equation $x^2 + y^2 + a^2 = 0$ where $a \in R$ may be

- (1) A point (2) A circle (3) A straight line (4) No real locus

Sol. Answer (1, 4)

Case-I : When $a \in R - \{0\}$

$\Rightarrow a^2$ is positive

$$x^2 + y^2 + a^2 = 0$$

So no real locus

Case-II : When $a = 0$

$$\Rightarrow x^2 + y^2 = 0$$

$$\Rightarrow x = 0; y = 0$$

Which is a point (0, 0)

20. The area formed by the straight lines represented by the equation $x^3 + y^3 - kx^2y + ay^2x = 0$ will be zero for

- (1) $k = 2$ (2) $k = 3$ (3) $k = 4$ (4) $k = 5$

Sol. Answer (1, 2, 3, 4)

If at all the equation,

$x^3 + y^3 - kx^2y + ay^2x = 0$ represents three straight lines then they all must pass through origin. So irrespective of k we choose the area will always remain zero.

21. A right angled triangle ABC , right angled at C , is such that $AC = a$; $BC = b$ and the angular points A and B slide along two given perpendicular axes respectively, the locus of C may be

- (1) $y = \frac{b}{a}x$ (2) $y = -\frac{b}{a}x$ (3) $y = \frac{a}{b}x$ (4) $y = -\frac{a}{b}x$

Sol. Answer (3, 4)

Let the co-ordinates of C be (h, k)

Now since $\angle AOB = 90^\circ = \angle ACB$

So, $OACB$ is a cyclic quadrilateral

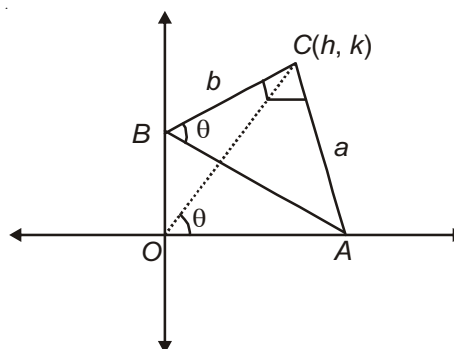
$$\Rightarrow \angle AOC = \angle ABC$$

$$\Rightarrow \tan \angle ABC = \frac{a}{b}$$

$$\Rightarrow \tan \theta = \frac{a}{b} = \frac{k}{h} \Rightarrow y = \frac{a}{b}x$$

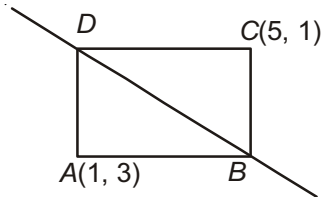
The required locus

Similarly the other locus may be $y = -\frac{a}{b}x$



22. The points (1, 3) and (5, 1) are two opposite vertices of a rectangle. The other two vertices lie on the line $y = 2x + c$, then
- (1) $c = 3$ (2) Other two vertices (4, 4) and (2, 0)
- (3) $c = -4$ (4) Other two vertices are (-4, 3) and (2, 3)

Sol. Answer (2, 3)



SECTION - C

Linked Comprehension Type Questions

Comprehension-I

a and b are real numbers between 0 and 1. $A(a, 1)$, $B(1, b)$ and $C(0, 0)$ are the vertices of a triangle.

1. If $\triangle ABC$ is equilateral, its area is

- (1) $\frac{7\sqrt{3} + 12}{4}$ (2) $2\sqrt{3} - 3$ (3) $\frac{\sqrt{3}}{4}$ (4) $\frac{\sqrt{3}}{2}$

Sol. Answer (2)

$A(a, 1)$; $B(1, b)$; $C(0, 0)$

$$(CA)^2 = (CB)^2 = (AB)^2$$

$$a^2 + 1 = b^2 + 1 = (a - 1)^2 + (b - 1)^2$$

$$a = b$$

$$b^2 + 1 = a^2 + b^2 - 2a - 2b + 2$$

$$0 = a^2 - 2a - 2b + 1$$

$$a^2 - 4a + 1 = 0$$

$$a = 2 - \sqrt{3}$$

$$\text{Side } CA = \sqrt{a^2 + 1} = \sqrt{8 - 4\sqrt{3}}$$

Area of equilateral,

$$\Delta = \frac{\sqrt{3}}{4} (\text{Side})^2$$

$$= \frac{\sqrt{3}}{4} (8 - 4\sqrt{3})$$

$$= \sqrt{3}(2 - \sqrt{3})$$

$$= 2\sqrt{3} - 3$$

2. If $\triangle ABC$ is isosceles with $AC = BC$ and $5(AB)^2 = 2(AC)^2$ then

(1) $ab = \frac{1}{4}$

(2) $ab = \frac{1}{8}$

(3) $ab = \frac{1}{16}$

(4) $ab = \frac{1}{2}$

Sol. Answer (1)

$$AC^2 = BC^2$$

$$\therefore a = b$$

$$5AB^2 = 2AC^2$$

$$5[(a-1)^2 + (b-1)^2] = 2(a^2 + 1)$$

$$5.2(a-1)^2 = 2(a^2 + 1)$$

$$2a^2 - 5a + 2 = 0$$

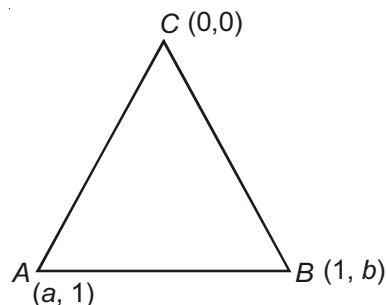
$$a = 2 \text{ \& } \frac{1}{2}$$

$$a < 1$$

$$\therefore a \in (0, 1) \text{ and } b \in (0, 1)$$

$$\therefore a = \frac{1}{2} = b$$

$$\therefore ab = \frac{1}{4}$$



3. If $\angle C = 90^\circ$ then

(1) $a + b = 0$

(2) $a - b = 0$

(3) $a + b = 1$

(4) $a - b = 1$

Sol. Answer (1)

$$AB^2 = AC^2 + BC^2$$

$$(a-1)^2 + (b-1)^2 = a^2 + 1 + b^2 + 1$$

$$-2a - 2b = 0$$

$$\therefore a + b = 0$$

Comprehension-II

Let the equations of perpendicular bisectors of sides AB and AC of $\triangle ABC$ be $x + y = 3$ and $x - y = 1$ respectively. The vertex A is (0, 0)

1. The orthocentre of $\triangle ABC$ is

(1) (1, 1)

(2) (2, 1)

(3) (0, 0)

(4) (0, 1)

Sol. Answer (3)

$$\text{Slope of } MP = -1$$

$$\therefore \text{Slope of } AB = 1$$

$$\text{Equation of } AB, y = x$$

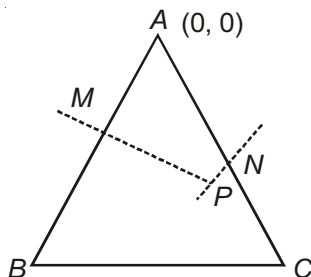
$$\text{Slope of } NP = 1$$

$$\therefore \text{Slope of } AC = -1$$

$$\text{Equation of } AC, y = -x$$

$$AB \perp AC$$

$$\therefore \text{orthocenter is } A(0, 0)$$



2. The circumcentre of the $\triangle ABC$ is

(1) (1, 1)

(2) (2, 1)

(3) (0, 0)

(4) (0, 1)

Sol. Answer (2)

Circumcentre is intersection point of $x + y = 3$ and $x - y = 1$ is $P(2, 1)$

3. Length of side BC of the $\triangle ABC$ is

(1) $\sqrt{2}$

(2) $\sqrt{8}$

(3) $\sqrt{12}$

(4) $\sqrt{20}$

Sol. Answer (4)

BC = Diameter of circumcircle of ABC

$$= 2 \text{ (radius of circle)} = 2 AP = 2 \sqrt{4+1} = 2\sqrt{5} = \sqrt{20}$$

Comprehension-III

Two adjacent sides of a parallelogram are $4x + 5y = 0$, and $7x + 2y = 0$.

1. If equation to one of the diagonals is $11x + 7y = 9$ then equation of the other diagonal is

(1) $x - y = 0$

(2) $7x - 11y = 0$

(3) $x + y = 0$

(4) $11x + 7y = 0$

Sol. Answer (1)

Equation of $OA \rightarrow 4x + 5y = 0$

...(i)

Equation of $OC \rightarrow 7x + 2y = 0$

...(ii)

Equation of $AC \rightarrow 11x + 7y = 9$

...(iii)

Solve equation (i) & (iii) $A\left(\frac{5}{3}, -\frac{4}{3}\right)$

Solve equation (ii) & (iii) $C\left(-\frac{2}{3}, \frac{7}{3}\right)$

Mid point of AC , $\left(\frac{1}{2}, \frac{1}{2}\right)$

$\therefore$ Point $B(1, 1)$ $\therefore$ Equation of OB , $y = x$

2. Vertices of the parallelogram are

(1) (0, 0)

(2) (1, 1)

(3) $\left(\frac{5}{3}, -\frac{4}{3}\right)$

(4) All of these

Sol. Answer (4)

Vertices are (0, 0), (1, 1), $\left(\frac{5}{3}, -\frac{4}{3}\right)$

3. Area of the parallelogram is

(1) 3

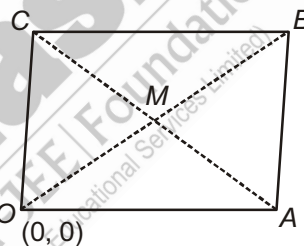
(2) $\frac{3}{2}$

(3) 6

(4) $\frac{2\sqrt{85}}{3}$

Sol. Answer (1)

$$\text{Area of parallelogram} = 2 [\text{Area of } \triangle OAB] = 2 \cdot \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ \frac{5}{3} & -\frac{4}{3} & 1 \\ 1 & 1 & 1 \end{vmatrix} = 3$$



Comprehension-IV

$A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are the vertices of a triangle ABC . $ax + by + c = 0$ is an equation of the line L .

1. If L intersects the sides BC , CA and AB of the $\triangle ABC$ at P , Q , R respectively then $\frac{BP}{PC} \times \frac{CQ}{QA} \times \frac{AR}{RB}$ is equal to

- (1) 1 (2) $-\frac{1}{2}$ (3) $\frac{1}{2}$ (4) -1

Sol. Answer (1)

Let $L \equiv ax + by + c = 0$

Then,

$$\frac{BP}{PC} \times \frac{CQ}{QA} \times \frac{AR}{RB} = \frac{|ax_2 + by_2 + c|}{|ax_3 + by_3 + c|} \times \frac{|ax_3 + by_3 + c|}{|ax_1 + by_1 + c|} \times \frac{|ax_1 + by_1 + c|}{|ax_2 + by_2 + c|} = 1$$

2. If the centroid of the $\triangle ABC$ is at the origin and algebraic sum of the lengths of the perpendiculars from the vertices of the triangle ABC on L is equal to 1 then sum of the squares of reciprocals of the intercepts made by L on the co-ordinate axes is equal to

- (1) 0 (2) 4 (3) 9 (4) 16

Sol. Answer (3)

$$\frac{x_1 + x_2 + x_3}{3} = 0 \Rightarrow x_1 + x_2 + x_3 = 0$$

Similarly $y_1 + y_2 + y_3 = 0$, let the line is $ax + by + c = 0$

$$\text{Then, } \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} + \frac{ax_2 + by_2 + c}{\sqrt{a^2 + b^2}} + \frac{ax_3 + by_3 + c}{\sqrt{a^2 + b^2}} = 1$$

$$a(x_1 + x_2 + x_3) + b(y_1 + y_2 + y_3) + 3c = \sqrt{a^2 + b^2}$$

$$3c = \sqrt{a^2 + b^2}$$

$$9c^2 = a^2 + b^2$$

$$\Rightarrow 9 = \frac{a^2}{c^2} + \frac{b^2}{c^2}$$

3. If P divides BC in the ratio 2 : 1 and Q divides CA in the ratio 1 : 3 then R divides AB in the ratio

- (1) 2 : 3 internally (2) 2 : 3 externally (3) 3 : 2 internally (4) 3 : 2 externally

Sol. Answer (4)

From Q. (1)

$$\frac{BP}{PC} \times \frac{CQ}{QA} \times \frac{AR}{RB} = 1$$

$$\frac{2}{1} \times \frac{1}{3} \times \frac{AR}{RB} = -1 \Rightarrow \frac{AR}{RB} = \frac{-3}{2}$$

$\Rightarrow R$ divides AB externally in the ratio 3 : 2

Comprehension-V

If α, β are roots of $x^2 - 6p_1x + 2 = 0$; β, γ are roots of $x^2 - 6p_2x + 3 = 0$ and γ, α are roots of equation $x^2 - 6p_3x + 6 = 0$ where p_1, p_2, p_3 are positive then

- The values of α, β, γ respectively are
 (1) 2, 3, 1 (2) 2, 1, 3 (3) 1, 2, 3 (4) -1, -2, -3
- The values of p_1, p_2, p_3 respectively are
 (1) $\frac{1}{2}, \frac{2}{3}, \frac{5}{6}$ (2) 1, 2, 5 (3) 6, 1, 4 (4) $2, \frac{3}{2}, \frac{6}{5}$
- If $A\left(\alpha, \frac{1}{\alpha}\right), B\left(\beta, \frac{1}{\beta}\right), C\left(\gamma, \frac{1}{\gamma}\right)$ be vertices of $\triangle ABC$ then centroid of $\triangle ABC$ is
 (1) $\left(4, \frac{11}{18}\right)$ (2) $\left(2, \frac{11}{18}\right)$ (3) $\left(-2, \frac{11}{18}\right)$ (4) $\left(-4, \frac{11}{18}\right)$

Solution of Comprehension-V

- Answer (2)
- Answer (1)
- Answer (2)

As α, β are roots of $x^2 - 6p_1x + 2 = 0$

$$\Rightarrow \alpha + \beta = 6p_1, \alpha\beta = 2 \quad \dots(i)$$

Also β, α, γ are roots of $x^2 - 6p_2x + 3 = 0$

$$\Rightarrow \beta + \gamma = 6p_2, \beta\gamma = 3 \quad \dots(ii)$$

And γ, α are roots of $x^2 - 6p_3x + 6 = 0$

$$\Rightarrow \gamma + \alpha = 6p_3, \gamma\alpha = 6 \quad \dots(iii)$$

$$\therefore (\alpha\beta) \cdot (\beta\gamma) \cdot (\gamma\alpha) = 2 \cdot 3 \cdot 6$$

$$\Rightarrow \alpha\beta\gamma = 6 \quad \dots(iv)$$

From (i), (ii), (iii), (iv) we get $\gamma = 3, \alpha = 2, \beta = 1$

$$\text{And } p_1 = \frac{1}{2}, p_2 = \frac{2}{3}, p_3 = \frac{5}{6}$$

Now, centroid of $\triangle ABC$, is

$$\left(\frac{\alpha + \beta + \gamma}{3}, \frac{1}{3} \left\{ \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} \right\} \right) \Rightarrow \left(\frac{6}{3}, \frac{1}{3} \left\{ \frac{1}{2} + 1 + \frac{1}{3} \right\} \right)$$

$$\Rightarrow \left(2, \frac{11}{18} \right)$$

Comprehension-VI

Through a given point O a straight line is drawn to cut two given straight lines in L and M having equations $a_1x + b_1y + 1 = 0$ and $a_2x + b_2y + 1 = 0$. Then locus of a point N on a variable line in each of following questions.

1. When ON is arithmetic mean of OL and OM

$$(1) 2a_1a_2x^2 + 2b_1b_2y^2 + 2\{a_1b_2 + a_2b_1\}xy - (a_1 + a_2)x - (b_1 + b_2)y = 0$$

$$(2) 2b_1b_2x^2 + 2a_1a_2y^2 + 2\{a_1b_2 + a_2b_1\}xy - (a_1 + a_2)x - (b_1 + b_2)y = 0$$

$$(3) 2b_1b_2x^2 + 2a_1a_2y^2 - 2\{a_1b_2 + a_2b_1\}xy - (a_1 + a_2)x - (b_1 + b_2)y = 0$$

$$(4) 2a_1a_2x^2 + 2b_1b_2y^2 - 2\{a_1b_2 + a_2b_1\}xy - (a_1 + a_2)x - (b_1 + b_2)y = 0$$

2. When ON is geometric mean of OL and OM

$$(1) a_1a_2x^2 + b_1b_2y^2 = 1$$

$$(2) a_1a_2x^2 + b_1b_2y^2 + (a_1b_2 + a_2b_1)xy = 1$$

$$(3) a_1a_2x^2 + b_1b_2y^2 + \{a_1b_2 + a_2b_1\}xy = 0$$

$$(4) a_1a_2x^2 + b_1b_2y^2 = \{a_1b_2 + a_2b_1\}xy$$

3. When ON is harmonic mean of OL and OM

$$(1) (a_1 + a_2)x + (b_1 + b_2)y = 2$$

$$(2) (a_1 - a_2)x + (b_1 - b_2)y = 2$$

$$(3) (a_1 + a_2)x + (b_1 + b_2)y = 0$$

$$(4) (a_1 - a_2)x + (b_1 - b_2)y = 0$$

Solution of Comprehension-VI

1. Answer (1)

2. Answer (2)

3. Answer (1)

Let 'O' be origin and let $a_1x + b_1y = 1$ and $a_2x + b_2y = 1$ be two given straight lines equation of straight line passing through 'O'

$$\frac{x}{\cos\theta} = \frac{y}{\sin\theta} \quad \dots(i)$$

This line cuts $a_1x + b_1y = 1$ and $a_2x + b_2y = 1$ at L and M respectively. Let $OL = r_1$ and $OM = r_2$. Then

$$L \equiv (r_1\cos\theta, r_1\sin\theta) \text{ and } M \equiv (r_2\cos\theta, r_2\sin\theta)$$

Let $N(h, k)$ be a variable point in equation (i) such that

$$ON = r_3 \Rightarrow h = r_3\cos\theta$$

$$k = r_3\sin\theta$$

Since L, M lie on $a_1x + b_1y = 1$ and $a_2x + b_2y = 1$

$$\therefore r_1(a_1\cos\theta + b_1\sin\theta) = 1 \text{ and } r_2(a_2\cos\theta + b_2\sin\theta) = 1$$

$$\Rightarrow \frac{r_1}{r_3} = \frac{1}{a_1h + b_1k} \text{ and } \frac{r_2}{r_3} = \frac{1}{a_2h + b_2k} \quad \dots(ii)$$

Now,

When ON is AM of OL and OM

$$\Rightarrow r_3 = \frac{r_1 + r_2}{2} \Rightarrow \frac{r_1}{r_3} + \frac{r_2}{r_3} = 2$$

$$\therefore \frac{1}{a_1h + b_1k} + \frac{1}{a_2h + b_2k} = 2$$

$$(a_1 + a_2)h + (b_1 + b_2)k - (a_1 + a_2)h - (b_1 + b_2)k + 2(a_1b_2 + a_2b_1)hk = 0 \text{ is required locus}$$

When ON is geometric mean of OL and OM

$$\Rightarrow r_3^2 = r_1 r_2 \Rightarrow \frac{r_1}{r_3} \cdot \frac{r_2}{r_3} = 1$$

$$\text{or } \left\{ \frac{r_3}{r_1} \right\} \cdot \left\{ \frac{r_3}{r_2} \right\} = 1 \Rightarrow (a_1 h + b_1 k)(a_2 h + b_2 k) = 1$$

$$\Rightarrow a_1 a_2 x^2 + b_1 b_2 y^2 + \{a_1 b_2 + a_2 b_1\}xy = 1 \text{ is required locus}$$

When ON is harmonic mean of OL and OM ;

$$\Rightarrow \frac{2}{r_3} = \frac{1}{r_1} + \frac{1}{r_2}$$

$$\Rightarrow \frac{r_3}{r_1} + \frac{r_3}{r_2} = 2$$

$$\Rightarrow (a_1 h + b_1 k) + (a_2 h + b_2 k) = 2$$

$$\Rightarrow (a_1 + a_2)h + (b_1 + b_2)k = 2 \text{ is required locus}$$

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : The straight lines $2x + 3y + 5 = 0$ and $3x - 2y + 1 = 0$ are perpendicular to each other.

and

STATEMENT-2 : Two lines $y = m_1 x + c_1$ and $y = m_2 x + c_2$ where $m_1, m_2 \in R$ are perpendicular if $m_1 \cdot m_2 = -1$.

Sol. Answer (1)

Statement-1: Product of slopes = -1

Statement-2: $\because m_1, m_2 \in R \therefore$ for perpendicular lines $m_1 m_2 = -1$

2. STATEMENT-1 : The lines $a_1 x + b_1 y + c_1 = 0$, $a_2 x + b_2 y + c_2 = 0$, $a_3 x + b_3 y + c_3 = 0$ are concurrent if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

and

STATEMENT-2 : The area of the triangle formed by three concurrent lines is always zero.

Sol. Answer (2)

3. STATEMENT-1 : The lines $(a + b)x + (a - 2b)y = a$ are concurrent at the point $\left(\frac{2}{3}, \frac{1}{3}\right)$.

and

STATEMENT-2 : The lines $x + y - 1 = 0$ and $x - 2y = 0$ intersect at the point $\left(\frac{2}{3}, \frac{1}{3}\right)$.

Sol. Answer (1)

$$a(x + y - 1) + b(x - 2y) = 0$$

Intersection point of $x + y - 1 = 0$ and $x - 2y = 0$ is $\left(\frac{1}{3}, \frac{2}{3}\right)$

4. STATEMENT-1 : Each point on the line $y - x + 12 = 0$ is equidistant from the lines $4y + 3x - 12 = 0$, $3y + 4x - 24 = 0$.

and

STATEMENT-2 : The locus of a point which is equidistant from two given lines is the angle bisector of the two lines.

Sol. Answer (3)

$P(x, x - 12)$ lies on line (1). Distance of equidistant lines from P .

$$PM = \left| \frac{3x + 4(x - 12) - 12}{5} \right| = \left| \frac{7x - 60}{5} \right|$$

$$PN = \left| \frac{4x + 3(x - 12) - 24}{5} \right| = \left| \frac{7x - 60}{5} \right|$$

$$PM = PN$$

5. STATEMENT-1 : Let $y = mx$, $my + x = 0$ and $3x + 4y = 5$ be the sides of a triangle then the radius of circumcentre of the triangle will be 1.

and

STATEMENT-2 : In a right angle triangle the radius of circumcircle is half of the hypotenuse of the triangle.

Sol. Answer (4)

m is variable

$\therefore$ Radius of circumcircle are also variable

6. STATEMENT-1 : If p, x_1, x_2, x_3 and q, y_1, y_2, y_3 form two arithmetic progression with common differences a and b , the centroid of Δ formed by points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ lies on a straight line.

and

STATEMENT-2 : For any two arithmetic progressions having first term p and q respectively with common differences ' a ' and ' b ' respectively. The point (h, k) given by $h = \frac{x_1 + x_2 + \dots + x_n}{n}$ and $k = \frac{y_1 + y_2 + \dots + y_n}{n}$ always lies on line $b(x - p) = a(y - q)$ for all values of n .

Sol. Answer (1)

$\therefore p, x_1, x_2, \dots$ and $q, y_1, y_2, y_3, \dots$ are in AP with common difference ' a ' and ' b ' respectively

$\therefore x_i = p + ai$ and $y_i = q + ib$

$$\therefore h = \frac{x_1 + x_2 + \dots + x_n}{n} \text{ and } k = \frac{y_1 + y_2 + \dots + y_n}{n}$$

$$\Rightarrow nh = \sum_{i=1}^n x_i \text{ and } nk = \sum_{i=1}^n y_i$$

$$\Rightarrow nh = \sum_{i=1}^n (p + ia) \text{ and } nk = \sum_{i=1}^n (q + ib)$$

$$\Rightarrow nh = np + \frac{n(n+1)}{2}a \text{ and } nk = nq + \frac{n(n+1)b}{2}$$

$$\Rightarrow \frac{h-p}{a} = \frac{n+1}{2} = \frac{k-q}{b}$$

$$\Rightarrow \frac{h-p}{a} = \frac{k-q}{b}$$

$\Rightarrow$ Hence locus of (h, k) is

$$\Rightarrow b\{x-p\} = a\{k-q\}$$

Hence, statement-2 is true and

$\therefore$ For statement-1, $n = 3$

$\therefore$ Statement-1 is true and statement-2 is correct explanation for statement-1

7. STATEMENT-1 : The joint equation of lines $2x - y = 5$ and $x + 2y = 3$ is $2x^2 + 3xy + 2y^2 - 11x - 7y = -15$.

and

STATEMENT-2 : Every second degree equation

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ always represents a pair of straight lines.

Sol. Answer (3)

As $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents the general equation of second degree.

But it represents a pair of straight line, if

$$\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\text{Also, } (2x - y - 5)(x + 2y - 3) = 2x^2 + 3xy - 2y^2 - 11x - 7y + 15 = 0$$

Hence option (3) correct.

8. If lines represented by $x^2 - 2pxy - y^2 = 0$ are rotated about origin through an angle θ , one in clockwise direction and other in anti-clockwise direction.

STATEMENT-1 : The equation of bisectors of angle between the lines in new position is $px^2 - 2xy + py^2 = 0$.

and

STATEMENT-2 : The bisectors of angles between lines in a new position are same as bisectors of angles between their old positions.

Sol. Answer (4)

The bisectors of angles between lines in new position are same as bisectors of angles between their old positions, is always true.

$\therefore$ The equation of angle bisectors of angle between lines in new position is

$$\frac{x^2 - y^2}{1 - (-1)} = \frac{xy}{-p} \Rightarrow px^2 + 2xy - py^2 = 0$$

Hence option (3) correct

9. STATEMENT-1 : The line $2009x + 2010y + 2011 = 0$ where $2009a + 2010b + 2011c = 0$ passes through the point $\left(\frac{a}{c}, \frac{b}{c}\right)$, where $abc \neq 0$.

and

STATEMENT-2 : If $2009a + 2010b + 2011c = 0$, then the straight line $ax + by + c = 0$, $abc \neq 0$ passes through $\left(\frac{2009}{2011}, \frac{2010}{2011}\right)$.

Sol. Answer (2)

10. STATEMENT-1 : The straight line $1005x - 2011y + 1006 = 0$ passes through $(1, 1)$.

and

STATEMENT-2 : The straight line $ax - by + c = 0$ passes through $(1, 1)$ if $a + c = b$.

Sol. Answer (1)

11. Lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q , respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R . **[IIT-JEE 2007]**

STATEMENT-1 : The ratio $PR : RQ$ equals $2\sqrt{2} : \sqrt{5}$.

and

STATEMENT-2 : In any triangle, bisector of an angle divides the triangle into two similar triangles.

Sol. Answer (3)

Angle bisector of a triangle does not divide the triangle into two similar triangles and hence statement-2 is wrong.

SECTION - E

Matrix-Match Type Questions

1. Match the mirror image of $(3, 2)$ with respect to the line in column-I

Column-I

- (A) $y = 0$
- (B) $x = 0$
- (C) $y = x$
- (D) $x + y = 0$

Column-II

- (p) $(-3, 2)$
- (q) $(3, -2)$
- (r) $(-2, -3)$
- (s) $(2, 3)$

Sol. Answer A(q), B(p), C(s), D(r)

2. The vertices of a triangle are $A(a, 0)$, $B(0, b)$ and $C(a, b)$

Column-I

- (A) Centroid
- (B) Circumcentre
- (C) Orthocentre
- (D) Foot of the altitude from C on AB

Column-II

- (p) $\left(\frac{a}{2}, \frac{b}{2}\right)$
- (q) (a, b)
- (r) $\left[\frac{a^3}{a^2 + b^2}, \frac{b^3}{a^2 + b^2}\right]$
- (s) $\left(\frac{2a}{3}, \frac{2b}{3}\right)$

Sol. Answer A(s), B(p), C(q), D(r)

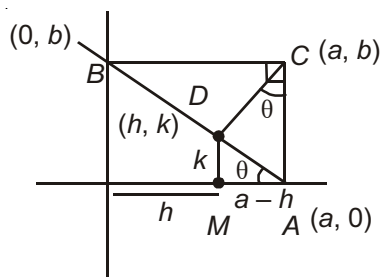
Centroid $\left(\frac{2a}{3}, \frac{2b}{3}\right)$

Circumcentre is middle point of $AB = \left(\frac{a}{2}, \frac{b}{2}\right)$

Orthocenter = (a, b)

Let foot of altitude from C is $D(h, k)$

$AD = AC \cdot \sin \theta = b \sin \theta$



$$DM = k = AD \sin \theta = b \sin^2 \theta \quad (\because \sin \theta = \frac{b}{\sqrt{a^2 + b^2}}, \cos \theta = \frac{a}{\sqrt{a^2 + b^2}} = \frac{b \cdot b^2}{a^2 + b^2} = \frac{b^3}{a^2 + b^2})$$

$$AM = a - h = AD \cos \theta$$

$$A - h = b \sin \theta \cos \theta = \frac{b \cdot b}{\sqrt{a^2 + b^2}} \cdot \frac{a}{\sqrt{a^2 + b^2}}$$

$$a - h = \frac{a \cdot b^2}{a^2 + b^2} \Rightarrow h = a \cdot \left(1 - \frac{b^2}{a^2 + b^2}\right) = \frac{a^3}{a^2 + b^2}$$

3. ABCD is a rectangle in the clock-wise direction. The co-ordinates of A are (1, 3) and of C are (5, 1) vertices B and D lie on line $y = 2x + c$. The co-ordinate of the

Column-I

(A) Middle point of BD

(B) Middle point of AB

(C) Middle point of BC

(D) Point D

Column-II

(p) (2, 0)

(q) (3, 2)

(r) $\left(\frac{5}{2}, \frac{7}{2}\right)$ (s) $\left(\frac{9}{2}, \frac{5}{2}\right)$ **Sol.** Answer A(q), B(r), C(s), D(p)Mid point of AC, $M(3, 2)$ M lie on $y = 2x + C$

$$\therefore C = -4$$

Equation of BD

$$y = 2x - 4$$

$$BM = r = AM = \sqrt{4+1} = \sqrt{5}$$

Slope of Line BD is $\tan \theta = 2$

$$\therefore \sin \theta = \frac{2}{\sqrt{5}}$$

$$\cos \theta = \frac{1}{\sqrt{5}}$$

Let B (h, k)

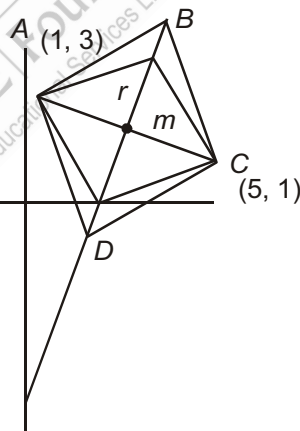
$$h = 3 \pm \sqrt{5} \cos \theta = 3 \pm 1 = 4, 2$$

$$k = 2 \pm \sin \theta = 2 \pm 2 = 4, 0$$

B (4, 4) & D (2, 0)

$$\text{Mid point of AB} = \left(\frac{5}{2}, \frac{7}{2}\right)$$

$$\text{Middle point of BC} = \left(\frac{9}{2}, \frac{5}{2}\right)$$



4. Reflection of the line $x + y + 1 = 0$ in the line

Column-I

- (A) $2x + y + 1 = 0$
 (B) $x - 2y + 1 = 0$
 (C) $x + 2y - 1 = 0$
 (D) $2x + y - 1 = 0$

Column-II

- (p) $x + 7y - 11 = 0$
 (q) $7x + y + 1 = 0$
 (r) $7x + y - 11 = 0$
 (s) $7x + y + 7 = 0$

Sol. Answer A(q), B(s), C(p), D(r)

5. Match column-I to column-II according to the given condition

Column-I

- (A) If the algebraic sum of perpendiculars from the points $(2, 3)$, $(1, 3)$ and $(3, 6)$ to a variable line is zero then the coordinates of the fixed point through which the lines passes are not equal to
 (B) For a triangle the coordinates of circum-centre and orthocentre are $(1, 1)$ and $(3, 3)$ respectively then the coordinates of centroid of the triangle are
 (C) The incentre of the triangle with vertices $(3, 0)$, $(0, 4)$ and $(0, 0)$ is
 (D) Orthocentre of the triangle formed by the lines $xy = 0$ and $x + y = 2$ is

Column-II

- (p) $\left(\frac{5}{3}, \frac{5}{3}\right)$
 (q) $(1, 1)$
 (r) $(0, 0)$
 (s) $(3, 4)$
 (t) $(1, 4)$

Sol. Answer A(p, q, r, s, t), B(p), C(q), D(r)

- (A) Let the line is $ax + by + c = 0$

and the points are (x_i, y_i) where $i = 1, 2, 3, \dots, n$. Algebraic length of perpendicular from (x_i, y_i)

$$p_i = \frac{ax_i + by_i + c}{\sqrt{a^2 + b^2}}$$

$$\text{But } \sum p_i = 0 \Rightarrow \frac{a \sum x_i + b \sum y_i + nc}{\sqrt{a^2 + b^2}} = 0$$

$$\Rightarrow \frac{a \sum x_i}{n} + \frac{b \sum y_i}{n} + c = 0 \quad \dots \text{(ii)}$$

by (i) and (ii)

$$x = \frac{\sum x_i}{n}, y = \frac{\sum y_i}{n}, \text{ which are the coordinates of fixed point}$$

According to the problem

$$x = \frac{2+1+3}{3} = 2$$

$$y = \frac{3+3+6}{3} = 4$$

Hence fixed point = $(2, 4)$

- (B) $\begin{array}{ccccc} (3, 3) & 2 & (x, y) & 1 & (1, 1) \\ \text{(Orthocenter)} & & \text{(centroid)} & & \text{(circumcentre)} \end{array}$

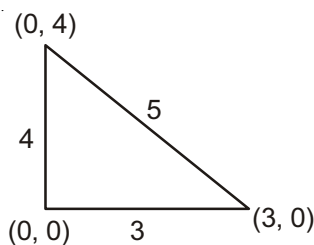
As we know that centroid divides the distance between orthocenter and circumcentre in the ratio 2 : 1.

$$\text{Hence, } x = \frac{2 \times 1 + 1 \times 3}{2 + 1} = \frac{5}{3}$$

$$y = \frac{2 \times 1 + 1 \times 3}{2 + 1} = \frac{5}{3}$$

Hence coordinates are $\left(\frac{5}{3}, \frac{5}{3}\right)$

- (C) Using the formula that coordinates of incentre are $\left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c}\right)$



According to the problem the incentre is given by

$$\left(\frac{5 \times 0 + 4 \times 3 + 3 \times 0}{4 + 3 + 5}, \frac{5 \times 0 + 4 \times 0 + 3 \times 4}{4 + 3 + 5}\right) = \left(\frac{12}{12}, \frac{12}{12}\right) = (1, 1)$$

- (D) Lines are $x = 0$, $y = 0$, $x + y = 2$. As we know that the lines $x = 0$ and $y = 0$ are perpendicular hence orthocenter is $(0, 0)$.
6. Consider a triangle ABC and $A = (2, 3)$. If the equation of angle bisectors of angles B and C is $y = x$ and $2x + 3y + 13 = 0$ respectively. Now match column-I and column-II according to the given condition.

Column-I

- (A) The equation of side BC is (p)
 (B) The coordinates of B are
 (C) The coordinates of C are
 (D) The coordinates of incentre of triangle

ABC are

Column-II

- (p) $\left(\frac{15}{2}, \frac{15}{2}\right)$
 (q) $11x - 9y - 15 = 0$
 (r) $\left(-\frac{24}{17}, -\frac{173}{51}\right)$
 (s) $\left(-\frac{13}{5}, -\frac{13}{5}\right)$
 (t) Is not parallel to $x + y + 1 = 0$

Sol. Answer A(q, t), B(p), C(r), D(s)

- (A) As we know that image of point A through $y = x$ and $2x + 3y + 13 = 0$ will lie on BC .

$$\text{Clearly } B_1 = (x_1, y_1) = (3, 2)$$

$$C_1 = (x_2, y_2) \text{ and the line mirror } 2x + 3y + 13 = 0.$$

To find the (x_2, y_2) we use the formula for image

$$\frac{x_2 - 2}{2} = \frac{y_2 - 3}{3} = \frac{-2(2 \times 2 + 3 \times 3 + 13)}{4 + 9}$$

$$\Rightarrow \frac{x_2 - 2}{2} = \frac{y_2 - 3}{3} = \frac{-2 \times 26}{13} = -4$$

$$\Rightarrow x_2 = -8 + 2, y_2 = -12 + 3$$

$$x_2 = -6, y_2 = -9$$

$$\Rightarrow C_1 = (-6, -9)$$

Hence equation of BC is $y - 2 = \frac{-9 - 2}{-6 - 3}(x - 3)$

$$\Rightarrow y - 2 = \frac{11}{9}(x - 3)$$

$$9y - 18 = 11x - 33 \Rightarrow 11x - 9y - 15 = 0$$

(B) B is the point of intersection of $y = x$ and $11x - 9y - 15 = 0$

$$\Rightarrow B = \left(\frac{15}{2}, \frac{15}{2}\right)$$

(C) Point ' C ' is intersection of

$$2x + 3y + 13 = 0 \quad \dots(i)$$

$$\text{and } 11x - 9y - 15 = 0 \quad \dots(ii)$$

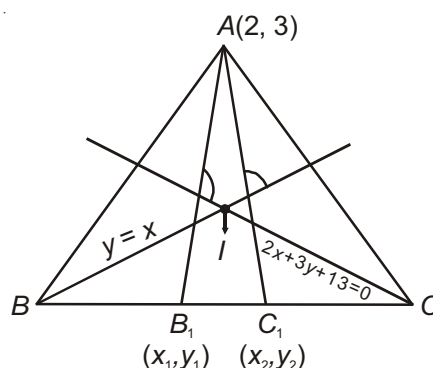
$$\Rightarrow C = \left(\frac{-24}{17}, \frac{-173}{51}\right)$$

(D) Incentre is the point of intersection of any two internal angle bisectors

$$y = x \quad \dots(i)$$

$$2x + 3y + 13 = 0 \quad \dots(ii)$$

$$\Rightarrow I = \text{incentre} = \left(-\frac{13}{5}, -\frac{13}{5}\right)$$



7. Match the reflection of the curve represented by $ax^2 + 2hxy + by^2 = 0$ about the line mentioned in column-I to the corresponding entries of column-II

Column-I

- (A) x-axis
(B) y-axis
(C) $x - y = 0$
(D) $x + y = 0$

Column-II

- (p) $bx^2 - 2hxy - ay^2 = 0$
(q) $bx^2 + 2hxy + ay^2 = 0$
(r) $ax^2 + 2hxy - by^2 = 0$
(s) $ax^2 - 2hxy + by^2 = 0$
(t) $bx^2 + 2hxy - ay^2 = 0$

Sol. Answer A(s), B(s), C(q), D(q)

(A) Reflection about x-axis

Replace $x \rightarrow x, y \rightarrow -y$

$$ax^2 + 2h(x)(-y) + b(-y)^2 = 0$$

$$ax^2 - 2hxy + by^2 = 0$$

(B) Replace $x \rightarrow -x$, $y \rightarrow y$

$$ax^2 - 2hxy + 3y^2 = 0$$

(C) Replace $x \rightarrow y$ and $y \rightarrow x$

$$bx^2 + 2hxy + ay^2 = 0$$

(D) Replace $x \rightarrow -y$ and $y \rightarrow -x$

$$bx^2 + 2hxy + ay^2 = 0$$

8. Consider the lines given by

$$L_1 : x + 3y - 5 = 0$$

$$L_2 : 3x - ky - 1 = 0$$

$$L_3 : 5x + 2y - 12 = 0$$

Match the Statements/Expressions in Column I with the Statements/Expressions in Column II. [IIT-JEE 2008]

Column I(A) L_1, L_2, L_3 are concurrent, if(B) One of L_1, L_2, L_3 is parallel to at least one of the other two, if(C) L_1, L_2, L_3 form a triangle, if(D) L_1, L_2, L_3 do not form a triangle, if**Column II**(p) $k = -9$ (q) $k = -\frac{6}{5}$ (r) $k = \frac{5}{6}$ (s) $k = 5$ **Sol.** Answer A(s); B(p, q); C(r), D(p, q, s)(A) Solving L_1 and L_3

$$\frac{x}{-36+10} = \frac{y}{+12-25} = \frac{1}{2-15}$$

$$\therefore x = 2, y = 1$$

 L_1, L_2, L_3 are concurrent, if (2, 1) lies on L_2

$$\therefore 6 - k - 1 = 0 \Rightarrow k = 5$$

(B) Either L_1 is parallel to L_2 , or L_3 is parallel to L_2 ,

$$\text{then } \frac{1}{3} = \frac{3}{-k} \text{ or } \frac{3}{5} = \frac{-k}{2}$$

$$\Rightarrow k = -9 \text{ or } k = -\frac{6}{5}$$

(C) L_1, L_2, L_3 form a triangle, if they are not concurrent, or not parallel

$$\therefore k \neq 5, -9, -\frac{6}{5} \Rightarrow k = \frac{5}{6}$$

(D) L_1, L_2, L_3 do not form a triangle, if

$$k = 5, -9, -\frac{6}{5}$$

SECTION - F

Integer Answer Type Questions

1. If three non-collinear points are given in a plane then the number of points which can make parallelogram with these three points, is

Sol. Answer (3)

Let us consider a triangle ABC where PQR are middle points of sides.

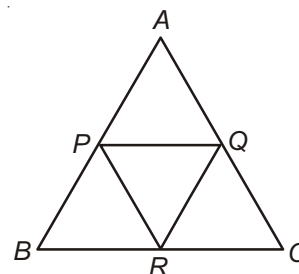
Here $PQ \parallel BC$ and $PQ = \frac{1}{2} BC$

Similarly $QR \parallel AB$ and $QR = \frac{1}{2} AB$

And $PR \parallel AC$ and $PR = \frac{1}{2} AC$

Hence quadrilaterals $PQCR$, $PQRB$, $PRQA$ are parallelogram.

So if three points P, Q, R are given then fourth point may have three positions A, B, C to form a parallelogram



2. The number of integral values of k if x coordinate of the point of intersection of the lines $5x + 7y = 12$ and $y = kx + 2$ is also an integer is

Sol. Answer (1)

$$5x + 7y = 12 \quad \dots(i)$$

$$y = kx + 2 \quad \dots(ii)$$

By (i) and (ii)

$$5x + 7(kx + 2) = 12$$

$$\Rightarrow x = \frac{-2}{5+7k}$$

Now for x to be integer

$$5 + 7k = 1, -1, 2, -2$$

$$\Rightarrow k = -\frac{4}{7}, -\frac{6}{7}, -\frac{3}{7}, -1$$

Hence only one value of k exists.

3. The number of lines passing through $(1, 1)$ and intersecting a segment of length 2 unit between the lines $x + y = 1$ and $x + y = 3$ is

Sol. Answer (2)

The equation of the given lines

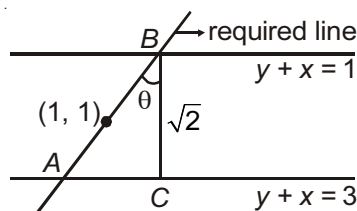
$$x + y = 1, x + y = 3$$

$$\text{Distance between the lines} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

The line passes through $(1, 1)$ intersects an intercepts length 2 unit which can be shown as following.

As $AB = 2$

$$\Rightarrow \cos \theta = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$



Let the slope of required line is m .

$$\tan 45^\circ = \pm \frac{m+1}{1-m} = \Rightarrow \pm \frac{m+1}{1-m} = 1, \frac{m+1}{1-m} = \pm 1$$

$$\text{If, } \frac{m+1}{1-m} = -1 \Rightarrow m+1 = m-1 \Rightarrow m = 0$$

$$\text{If, } \frac{m+1}{1-m} = 1 \Rightarrow m+1 = -1+m, \Rightarrow m \rightarrow \infty$$

Hence the equation of the lines is $y = 1$ and $x = 1$

Hence two lines are there.

4. The axes are rotated through an angle, keeping origin fixed then the equation of the line $3x + 4y - 12 = 0$ becomes $ax + by - ab = 0$. If $\frac{1}{a^2} + \frac{1}{b^2} = k^2$, then the value of $12k$ is

Sol. Answer (5)

If origin is fixed then the perpendicular distance of $(0, 0)$ from the line is always constant.

$$\Rightarrow \frac{1}{\sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{1}{4}\right)^2}} = \frac{1}{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2}$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{9} + \frac{1}{16} = \frac{25}{144} = \left(\frac{5}{12}\right)^2$$

$$\Rightarrow \frac{5}{12} = k \Rightarrow 12k = 5$$

5. The total number of straight lines that pass through origin and are at a distance of 3 units from the line $x + y = 6$ is

Sol. Answer (0)

The distance of the line $x + y = 6$ from origin $(0, 0)$ is

$$\left| \frac{-6}{\sqrt{2}} \right| = 3\sqrt{2}$$

So, the minimum distance of the line $x + y = 6$ from $(0, 0)$ is $3\sqrt{2}$, which is greater than the given distance, hence no such line is possible.

6. Let a line $\frac{x}{a} + \frac{y}{b} = 1$ intersects the x -axis at A and y -axis at B respectively. A line parallel to it is drawn to intersect the axes in A' and B' respectively. The extremities of the lines are joined transversely. If the locus of point of intersection of the line joining them is $\frac{x}{a} = c \frac{y}{b}$, then c is equal to

Sol. Answer (1)

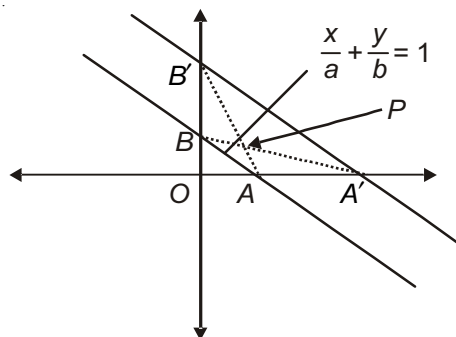
Let 'O' be the origin clearly $OA = a$; $OB = b$

Let the other line be $A'B'$

$$\text{Let } \frac{OA'}{OA} = \frac{OB'}{OB} = \lambda (\text{say})$$

So, the equation to AB' is

$$\frac{x}{a} + \frac{y}{\lambda b} = 1$$



And that to $A'B$ is

$$\frac{x}{\lambda a} + \frac{y}{b} = 1$$

Let (h, k) be their point of intersection

$$\frac{h}{a} + \frac{k}{\lambda b} = 1 \quad \dots(A)$$

$$\frac{h}{\lambda a} + \frac{k}{b} = 1 \quad \dots(B)$$

Subtracting (A) and (B) we get

$$\frac{h}{a} \left(1 - \frac{1}{\lambda}\right) - \frac{k}{b} \left(1 - \frac{1}{\lambda}\right) = 0$$

$$\Rightarrow \frac{h}{a} = \frac{k}{b}$$

So, the locus is

$$\frac{x}{a} = \frac{y}{b}$$

Hence $c = 1$

7. The lines joining the origin to the points of intersection of the straight line, $kx + hy = 2hk$ with the curve $(x - h)^2 + (y - k)^2 = c^2$ are at right angles, then $h^2 + k^2 = tc^2$, then t is equal to

Sol. Answer (1)

Making the equation

$$(x - h)^2 + (y - k)^2 = c^2 \text{ homogeneous}$$

with $kx + hy = 2hk$, we get the required condition that

$$\frac{kx + hy}{2hk} = 1$$

$$h^2 + k^2 = c^2$$

Hence $t = 1$

8. A piece of cheese is kept at $P(1, 2)$ in a plane. A rat sitting somewhere in the fourth quadrant with no sense of direction starts moving towards the cheese along the line $x - y = 3$, and after reaching a point $Q(a, b)$, it starts getting farther from the cheese, then $(a + b)$ is equal to

Sol. Answer (3)

Let the point Q be $(h + 3, h)$

Now since after reaching point Q it starts moving farther

So, PQ must be perpendicular to the line

$$\text{Slope of } PQ = \frac{h - 2}{h + 2}$$

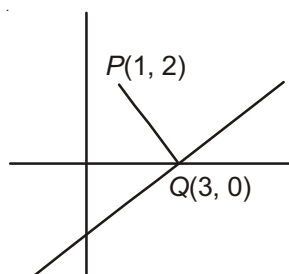
Slope of the line 1

$$\Rightarrow h - 2 = -(h + 2)$$

$$\Rightarrow 2h = 0 \Rightarrow h = 0$$

So, $Q = (3, 0)$

$\therefore a + b$ is 3



SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : If three points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) are collinear, then $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

STATEMENT-2 : If $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$, then the points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) will be collinear.

STATEMENT-3 : If lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are concurrent then

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

- (1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (2)

All the statements are clearly true.

2. STATEMENT-1 : If one angle between two intersecting line is θ then other angle will be $\pi - \theta$.

STATEMENT-2 : There is always two angles between two intersecting lines.

STATEMENT-3 : The equation of the line passes through $(2, 3)$ and which is farthest from the point $(0, 0)$ is $3y + 2x - 13 = 0$.

- (1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (2)

STATEMENT-1 is clearly true.

STATEMENT-2 is true because if lines intersect at right angle, then both angles are equal.

STATEMENT-3

Let $A = (2, 3)$, $B = (0, 0)$, the required line will be perpendicular to AB .

$$\text{Slope of } AB = \frac{3}{2}$$

Hence required line is

$$y - 3 = -\frac{2}{3}(x - 2)$$

$$3y - 9 = -2x + 4$$

$$3y + 2x - 13 = 0$$

3. Consider a triangle ABC in xy -plane with D , E and F as the middle points of the sides BC , CA and AB respectively. If the co-ordinates of the point D , E , F are $\left(\frac{3}{2}, \frac{3}{2}\right)$, $\left(\frac{7}{2}, 0\right)$ and $\left(0, \frac{-1}{2}\right)$ then :

STATEMENT-1 : Circumcentre of triangle ABC lies inside the triangle.

STATEMENT-2 : Orthocentre, centroid, circumcentre and incentre of triangle DEF are collinear but of triangle ABC are non-collinear.

STATEMENT-3 : Distance between centroid and orthocentre of $\triangle ABC$ is $\frac{5\sqrt{2}}{3}$.

(1) F F T

(2) T T F

(3) F T F

(4) T F T

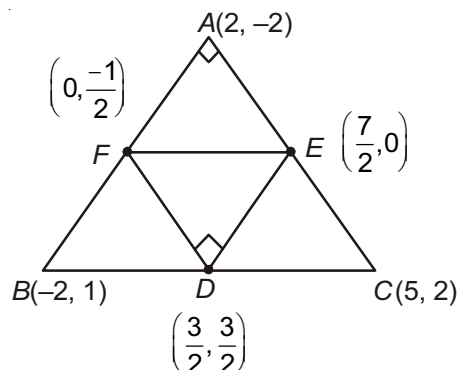
Sol. Answer (1)

$$m_1 = \text{slope of } FD = \frac{2}{3} \times 2 = \frac{4}{3}$$

$$m_2 = \text{slope of } DE = \frac{-\frac{3}{2}}{\frac{2}{1}} = -\frac{3}{4}$$

$m_1 m_2 = -1$. Hence $\angle D$ is 90° hence $\angle A$ is also 90° .

$\triangle ABC$ and $\triangle DEF$ are both right angled isosceles triangle.



4. STATEMENT-1 : One of the lines represented by the equation $x^3 + \lambda x^2 y + \lambda x y^2 + y^3 = 0$ will bisect the angle between the other two.

STATEMENT-2 : The equation $k(x^3 - 3xy^2) + y^3 - 3x^2y = 0$ represents three straight lines equally inclined.

STATEMENT-3 : The distance between the two parallel straight lines represented by $8x^2 + 8xy + 2y^2 + 26x + 13y + 15 = 0$ is $\frac{7}{2\sqrt{5}}$.

(1) T F T

(2) T T T

(3) F F T

(4) T T F

Sol. Answer (2)

1. Statement-1

$$x^3 + y^3 + \lambda xy(x + y) = 0$$

$$(x + y) \{x^2 + y^2 - xy - \lambda xy\} = 0$$

$$(x + y) (x^2 + y^2 + (\lambda - 1)xy) = 0$$

The angle bisector of lines $x^2 + y^2 + (\lambda - 1)xy = 0$

$$\frac{x^2 - y^2}{1 - 1} = \frac{2xy}{\lambda - 1} \Rightarrow (x - y)(x + y) = 0$$

Hence $x + y = 0$ is angle bisector of other two lines

2. Statement-2

Let $y = x \tan \theta$ be one of the lines represented by the equation

$$k(x^3 - 3xy^2) + y^3 - 3x^2y = 0$$

$$\text{Then, } \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} = k = \tan 3\alpha (\text{say}) \Rightarrow \tan 3\theta = \tan 3\alpha$$

$$\theta = \frac{1}{3} n\pi + \alpha, n \in \mathbb{I}; \text{ angle which the lines makes with } x \text{ axis are } \alpha, \frac{\pi}{3} + \alpha, \frac{2\pi}{3} + \alpha$$

$\therefore$ Angle between lines 1st and 2nd is $\frac{\pi}{3}$, between 2nd and 3rd is $\frac{\pi}{3}$, so between 1st and 3rd is $\frac{2\pi}{3}$

which is $\pi - \frac{2\pi}{3}$ i.e., $\frac{\pi}{3}$. Hence three lines equally inclined to one another.

3. Statement-3

Distance between two parallel lines is $2\sqrt{\frac{g^2 - ac}{a(a+b)}}$

Hence $a = 8, b = 2, h = 4, g = 13$

$$f = \frac{13}{2}, c = 15 \Rightarrow 2\sqrt{\frac{169 - 120}{8 \times 10}} = 2\sqrt{\frac{49}{8 \times 10}} = \frac{7}{2\sqrt{5}}$$

5. STATEMENT-1 : Let $\triangle ABC$ be right angle with vertices $A(0, 2), B(1, 0)$ and $C(0, 0)$. If D is the point on AB such that the segment CD bisects angle C then the length of CD is $\frac{2\sqrt{2}}{3}$.

STATEMENT-2 : The number of points on the straight line which joins $(-4, 11)$ to $(16, -1)$ whose co-ordinates are positive integer is 3.

STATEMENT-3 : If $k = 2$ then the lines $L_1 : 2x + y - 3 = 0, L_2 : 5x + ky - 3 = 0$ and $L_3 : 3x - y - 2 = 0$ are concurrent.

(1) F T F

(2) T F T

(3) T T F

(4) F F T

Sol. Answer (3)

Statement-1

Area of $\triangle ACD$ + area of $\triangle BCD$ = Area of $\triangle ABC$

$$\therefore \frac{1}{2} \times 1 \times x \sin 45^\circ + \frac{1}{2} \times x \times 2 \sin 45^\circ = \frac{1}{2} \times 1 \times 2$$

$$3x \sin 45^\circ = 2 \Rightarrow x = \frac{2\sqrt{2}}{3}$$

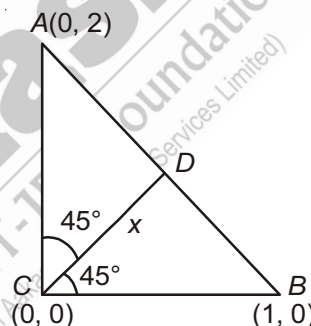
Statement-2: Slope of the line = $\frac{3}{5}$

Equation of the line is $\Rightarrow 3x + 5y = 43$

$$5y = 43 - 3x \Rightarrow y = \frac{43 - 3x}{5}$$

Hence the points are $[(1, 8), (6, 5), (11, 2)]$

Statement-3: Here $\begin{vmatrix} 2 & 1 & -3 \\ 5 & k & -3 \\ 3 & -1 & -2 \end{vmatrix} = 0$. Hence $k = -2$.



6. STATEMENT-1 : A rectangle is sometimes a rhombus.

STATEMENT-2 : A rhombus is always a parallelogram.

STATEMENT-3 : The diagonals of a parallelogram always bisect the angles of the vertices.

(1) F T F

(2) F F T

(3) T F T

(4) T T F

Sol. Answer (4)

7. STATEMENT-1 : If the x intercept of the line $y = mx + 2$ is greater than $\frac{1}{2}$ then the gradient of line lies in the interval $(-4, 0)$

STATEMENT-2 : The equation of the line making equal intercepts on the axes and touching $|z|^2 = \frac{1}{2}$ is

$|x + y| = 1$, where $z = x + iy$.

STATEMENT-3 : The equation of the line passing through $(-4, -2)$ having equal intercept on the co-ordinate axes is $y = x$ and $x + y - 6 = 0$

(1) F T F

(2) F T T

(3) T T F

(4) F F T

Sol. Answer (3)

Statement-1

The x intercept of the line = $\frac{-2}{m}$

$$-\frac{2}{m} > \frac{1}{2} \Rightarrow \frac{1}{2} + \frac{2}{m} < 0 \Rightarrow \frac{m+4}{2m} < 0$$

$$m \in (-4, 0)$$

Statement-2

Let equation of line $x + y = a$... (i)

$$|z|^2 = \frac{1}{2} \Rightarrow x^2 + y^2 = \frac{1}{2} \quad \dots (ii)$$

$y = a - x$ putting in equation (ii) is

$$x^2 + (a - x)^2 = \frac{1}{2} \Rightarrow 2x^2 - 2ax + a^2 - \frac{1}{2} = 0$$

$$D = 0 \Rightarrow a^2 = 1 \Rightarrow a = \pm 1$$

Hence the equation of lines is $|x + y| = 1$

Statement-3

Let the equation of line is

$$y + 2 = m(x + 4) \Rightarrow mx - y = 2 - 4m$$

$$\text{or } \frac{x}{\frac{2-4m}{m}} + \frac{y}{4m-2} = 1$$

$$\therefore \left| \frac{2-4m}{m} \right| = |4m-2|$$

$$m = \frac{1}{2}, x - y + 2 = 0, 4m - 2 = 0 \text{ or } m = \pm 1$$

$$2y = x, x + y + 6 = 0, m = +1$$

SECTION - H

Aakash Challengers Questions

1. The lines $xy = 0$ and $x + y = 17$ form a triangle in the x - y plane. The total number of points having co-ordinates which are prime numbers and lie inside the triangle is

(1) 23 (2) 24 (3) 25 (4) 26

Sol. Answer (4)

Prime number between 0 & 17; 2, 3, 5, 7, 11, 13

With 2 as x -co-ordinate, total points = 6

With 3 as x -co-ordinate, total points = 6

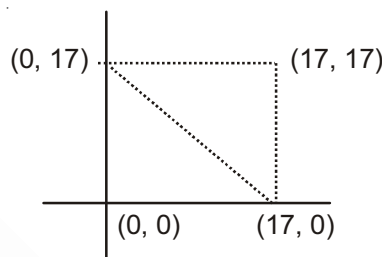
With 5 as x -co-ordinate, total points = 5

With 7 as x -co-ordinate, total points = 4

With 11 as x -co-ordinate, total points = 3

With 13 as x -co-ordinate, total points = 2

$\therefore$ Total points = 26



2. A line passes through $A(1, 1)$ and $B(100, 1000)$. The number of points with integral co-ordinates on the line strictly between A and B is

(1) 7 (2) 8 (3) 9 (4) 10

Sol. Answer (2)

The slope of the line is

$$\frac{1000 - 1}{100 - 1} = \frac{111}{11}$$

So, all the points will have the form $(1 + 11t, 1 + 111t)$

$$\Rightarrow 1 < 1 + 11t < 100 \quad | \quad 1 < 1 + 111t < 1000$$

$$\Rightarrow 0 < 11t < 99 \quad | \quad 0 < 111t < 999$$

$$\Rightarrow 0 < t < 9 \quad | \quad 0 < t < 9$$

Hence there are 8 such values of t and hence there are 8 such points.

3. A variable line passing through a fixed point P cuts the given curve $y^2 = 4ax$ at Q and R . Now a variable point S is chosen on this line in such a way that PQ , PS and PR are in H.P. Prove that the locus of S is a straight line whose slope is independent of the abscissa of the point P .

Sol. Let $P \equiv (\alpha, \beta)$ and any line through P be

$$\frac{x - \alpha}{\cos \theta} = \frac{y - \beta}{\sin \theta} = r$$

Any point on this line is $(\alpha + r \cos \theta, \beta + r \sin \theta)$

Points where it meets the parabola can be obtained from;

$$(\beta + r \sin \theta)^2 = 4a(\alpha + r \cos \theta)$$

$$\Rightarrow r^2 \sin^2 \theta + 2r(\beta \sin \theta - 2a \cos \theta) + \beta^2 - 4a\alpha = 0$$

$$\Rightarrow PQ + PR = \frac{2(2a \cos \theta - \beta \sin \theta)}{\sin^2 \theta} \text{ and } PQ \cdot PR = \frac{\beta^2 - 4a\alpha}{\sin^2 \theta}$$

Since PQ , PS and PR are in H.P.

$$\Rightarrow PS = \frac{(\beta^2 - 4a\alpha)}{(2a\cos\theta - \beta\sin\theta)}$$

Let $S \equiv (h, k) \Rightarrow h = \alpha + PS\cos\theta, k = \beta + PS\sin\theta$.

$$\Rightarrow \beta^2 - 4a\alpha = 2a(h - \alpha) - \beta(k - \beta)$$

Hence locus of 'S' is $2ax - y\beta + 2a\alpha = 0$

Which is clearly a straight line whose slope is $2a/\beta$, that does not depend upon the abscissa of point P.

4. In a $\triangle ABC$, $A \equiv (\alpha, \beta)$, $B \equiv (1, 2)$, $C \equiv (2, 3)$ and point 'A' lies on the line $y = 2x + 3$ where $\alpha, \beta \in \text{integer}$ and area of the triangle is S such that $[S] = 2$ {where $[.]$ denotes greatest integer function}. Possible co-ordinates of A may be

- (1) $(-7, -11)$ (2) $(-6, -9)$ (3) $(2, 7)$ (4) $(3, 9)$

Sol. Answer (1, 2, 3, 4)

$A(\alpha, \beta)$ lies on $y = 2x + 3$

$$\beta = 2\alpha + 3$$

$$A \equiv (\alpha, 2\alpha + 3)$$

$$\text{Area of } \Delta = \left| \frac{1}{2} \begin{vmatrix} \alpha & 2\alpha + 3 & 1 \\ 1 & 2 & 1 \\ 2 & 3 & 1 \end{vmatrix} \right|$$

$$= \left| \frac{1}{2} [\alpha(2-3) + (2\alpha+3)(2-1) + 1(3-4)] \right|$$

$$= \frac{1}{2} |\alpha + 2| = S$$

$$\Rightarrow 4 \leq |\alpha + 2| < 6$$

$$\Rightarrow 4 \leq \alpha + 2 < 6 \text{ or } -6 < \alpha + 2 \leq -4$$

$$2 \leq \alpha < 4 \text{ or } -8 < \alpha \leq -6$$

$\therefore$ Integral values of α are

$$2, 3, -6, -7$$

So, we get the co-ordinates as

$$(-7, -11), (-6, -9), (2, 7) \text{ and } (3, 9)$$

5. A line 'L' is drawn from $(4, 3)$ to meet the lines $L_1 : 3x + 4y + 5 = 0$ and $L_2 : 3x + 4y + 15 = 0$ at point A and B respectively. From 'A' a line, perpendicular to L is drawn meeting the line L_2 at A_1 . Similarly from point 'B' a line, perpendicular to L is drawn meeting the line L_1 at B_1 . Thus a parallelogram AA_1BB_1 is formed. The equation(s) of 'L' so that the area of parallelogram AA_1BB_1 is least, may be

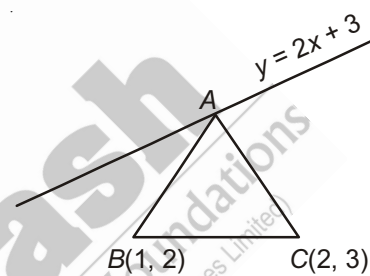
(1) $(y-3) = \frac{1}{7}(x-4)$ (2) $(y-3) = -7(x-4)$ (3) $(y-3) = \frac{1}{6}(x-4)$ (4) $(y-3) = -6(x-4)$

Sol. Answer (1, 2)

$$L_1 : 3x + 4y + 5 = 0$$

$$L_2 : 3x + 4y + 15 = 0$$

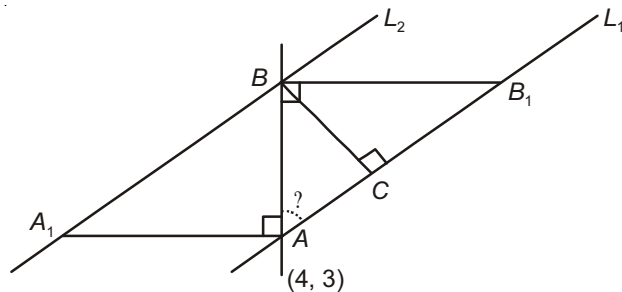
$$\text{Distance between the two lines} = \left| \frac{15-10}{\sqrt{3^2+4^2}} \right| = 2$$



$BC = 2$ units, let $\angle BAB_1 = \theta$

$AB = 2 \operatorname{cosec} \theta$, $BB_1 = 2 \sec \theta$

$$\begin{aligned} \Rightarrow \operatorname{ar}(\triangle ABB_1) &= \frac{1}{2} AB \cdot BB_1 \\ &= \frac{1}{2} \cdot 2 \operatorname{cosec} \theta \cdot 2 \sec \theta \\ &= \frac{4}{\sin 2\theta} \end{aligned}$$



Now area of parallelogram $AA_1 BB_1$

$$= 2 \triangle ABB_1 = \frac{8}{\sin 2\theta}$$

Clearly this is least when $\sin 2\theta = 1$

$$\Rightarrow \theta = \frac{\pi}{4}$$

Let the slope of the line drawn be 'm'

$$\Rightarrow 1 = \left| \frac{m + \frac{3}{4}}{1 - \frac{3m}{4}} \right| \Rightarrow m = -7, \frac{1}{7}$$

Hence the equation of lines

$$(y - 3) = \frac{1}{7}(x - 4) \text{ or } (y - 3) = -7(x - 4)$$

6. Let $P(\sin \theta, \cos \theta)$ ($0 \leq \theta \leq 2\pi$) be a point and let OAB be a triangle with vertices $(0, 0)$, $\left[\frac{\sqrt{3}}{2}, 0\right]$ and $\left[0, \frac{\sqrt{3}}{2}\right]$.

The range of θ if P lies inside the triangle OAB may be

- (1) $\left(0, \frac{\pi}{12}\right)$ (2) $\left(\frac{5\pi}{12}, \frac{\pi}{2}\right)$ (3) $\left[\pi, \frac{5\pi}{3}\right]$ (4) $\left[\frac{\pi}{2}, \pi\right]$

Sol. Answer (1, 2)

Equation of lines along OA , OB and AB are $y = 0$, $x = 0$, and $x + y = \frac{\sqrt{3}}{2}$ respectively. Now P and B will lie on the same side of $y = 0$ if $\cos \theta > 0$. Similarly P and A will lie on the same side of $x = 0$ if $\sin \theta > 0$ and P and Q will lie on the same side of $x + y = \frac{\sqrt{3}}{2}$ if

$$\sin \theta + \cos \theta < \frac{\sqrt{3}}{2}$$

$$\text{Now, } \sin \theta + \cos \theta < \frac{\sqrt{3}}{2}$$

$$\Rightarrow \sin\left(\theta + \frac{\pi}{4}\right) < \frac{\sqrt{3}}{2}$$

$$\Rightarrow 0 < \theta + \frac{\pi}{4} < \frac{\pi}{3}$$

$$\Rightarrow \frac{2\pi}{3} < \theta + \frac{\pi}{4} < \pi$$

Since $\sin\theta > 0$ and $\cos\theta > 0$

$$\Rightarrow 0 < \theta < \frac{\pi}{12} \text{ or } \frac{5}{12}\pi < \theta < \frac{\pi}{2}$$

7. Two sides of a triangle have the joint equation $x^2 - 2xy - 3y^2 + 8y - 4 = 0$. The third side, which is variable, always passes through the point $(-5, -1)$. Find the range of values of the slope of the third side such that origin is an interior point.

Sol. Combined equation of line is

$$x^2 - 2xy - 3y^2 + 8y - 4 = 0$$

$$\Rightarrow (x - y)^2 = 4y^2 - 8y + 4$$

$$\Rightarrow x - y = \pm 2(y - 1)$$

Thus two sides of a triangle are

$$L_1 : 3y - x - 2 = 0 \text{ and } L_2 : y + x - 2 = 0$$

And these intersect at $A \equiv (1, 1)$

Let the third side be $(y + 1) = m(x + 5)$

$$\Rightarrow L_3 : y = mx + 5m - 1$$

Let L_3 meet the line L_1 and L_2 at B and C respectively

$$B \equiv \left(\frac{15m - 5}{1 - 3m}, -1 \right); C \equiv \left(\frac{3 - 5m}{1 + m}, \frac{7m - 1}{1 + m} \right)$$

Now as origin has to be an interior point so

$$(1 - 5m)(1 - m + 1 - 5m) > 0$$

$$\Rightarrow m > \frac{1}{3} \text{ or } m < \frac{1}{5}$$

Similarly, points 'O' and 'C' should be on the same side of line AB.

$$\Rightarrow -2 \left(\frac{3(7m - 1)}{1 + m} - \frac{3 - 5m}{1 + m} - 2 \right) > 0$$

$$\Rightarrow -1 < m < \frac{1}{3}$$

Finally points 'O' and B should lie on the same side of AC

$$-2 \left(\frac{15m - 5}{1 - 3m} + (-1) - 2 \right) > 0$$

$$\left(m - \frac{1}{3} \right)^2 > 0 \text{ and } m \in R$$

$$\Rightarrow m \in \left(-1, \frac{1}{5} \right)$$

8. If the line $y = 3x$ meets the lines $x = 1, x = 2, \dots, x = 12$ at points $A_1, A_2, \dots, A_{12}$ respectively then $(OA_1)^2 + (OA_2)^2 + \dots + (OA_{12})^2$ is equal to _____.

Sol. $A_n (n, 3n)$

$$(OA_n)^2 = n^2 + 9n^2 = 10n^2$$

$$\sum_{n=1}^{12} (OA_n)^2 = 10 \cdot \sum_{n=1}^{12} n^2 = \frac{10 \cdot (12+1)(24+1)12}{6} = 6500$$

9. $A_1, A_2, \dots, A_n$ are points on the line $y = x$ lying in the positive quadrant such that $OA_n = n$. OA_{n-1} . O being the origin. If $OA_1 = 1$ then the co-ordinates of A_8 are $(3a\sqrt{2}, 3a\sqrt{2})$ where a is equal to _____.

Sol. $OA_3 = 3$. $OA_2 = 3$. $2! = 3!$, $OA_4 = 4!$

Hence $OA_8 = 8!$

$$\sqrt{18a^2 + 18a^2} = 8!$$

$$6a = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$a = \pm 6720$$

10. The co-ordinates of a point A_n is $(n, n\sqrt{n})$ where $n \in N$. If $O(0, 0)$ then $\sum_{i=1}^{12} (OA_i)^2$ equals _____.

Sol. $OA_n^2 = n^2 + n^3$

$$\sum_{n=1}^{12} (OA_n)^2 = \frac{n(n+1)(2n+1)}{6} + \frac{n^2(n+1)^2}{4}$$

$$\text{Put } n = 12 = 6734$$

11. The straight lines $L : x + y + 1 = 0$ and $L_1 : x + 2y + 3 = 0$ are intersecting. m is the slope of the straight line L_2 such that L is the bisector of the angle between L_1 and L_2 . Then $812m^2 + 3$ equals _____.

Sol. $m_1 = -1$, $m_2 = -\frac{1}{2}$, $m_3 = m$

Line L is bisector of angle of other two.

$$\therefore \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{m_1 - m_3}{1 + m_1 m_3} \right|$$

$$\frac{1}{3} = \left| \frac{m+1}{m-1} \right|$$

$$\therefore m = -2$$

$$812(m^2) + 3 = 812(4) + 3 = 3251$$

